

# CONTEMPORARY ISSUES IN ECONOMICS AND FINANCE: THE VIEW FROM CHINA



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# Contemporary Issues in Economics and Finance: The View from China

Edited by Kenneth S. Bigel, Ph.D

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## Introduction

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This is a collection of research papers written almost exclusively by high school students with one exception where a paper was produced by a college student. All the students are Chinese, with some residing in the People's Republic of China (PRC) while others live here in North America. I have worked with these wonderful young people on these projects over the course of many months, starting with the selection of a topic of worthwhile inquiry and on through the many interim stages of Literature Review and data gathering to the completion of the exciting, finished products which you may read in the pages that follow.

These budding researchers all share a desire to learn and advance, and all display the work ethic required to do so. The reader will sense the struggles encountered in the conduct of research and writing, and the valuable lessons learned in the process. The research topics chosen represent what is on the minds of our friends on the other side of the world and here in North America, with the aspiration to build bridges between the two most powerful nations on Earth today.

In my regular university teaching, I have found that requiring young students to read published scholarly research in order to learn how to do their own explorations is a frail and deficient effort particularly in the short time available to secondary and undergraduate students. This compendium is intended, inter alia, to improve this shortcoming. It can be used as a medium for teaching undergraduate students how to conduct basic research in business and the social sciences while still in college. This collection provides a set of tools that, once learned, will carry the researcher throughout his/her future intellectual and professional life. Readers will view a variegated selection of topics and investigative approaches that are presented at a palatable academic standard. It exhibits the various research procedures that can be utilized and a wide variety of topics that may be accessed for a larger cohort of ambitious secondary and college students across our shared planet.

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## CHAPTER OVERVIEW

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- 1.1: Examples of Social Interdependencies in Microeconomic Theory
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## 1.1: Examples of Social Interdependencies in Microeconomic Theory

**Title:** Examples of Social Interdependencies in Microeconomic Theory

**Author:** Yunzhen Lawrence Ju

**Abstract:** Microeconomics is a social science that involves complicated interactions and interdependencies among individual people. This paper explores how microeconomics investigates social interdependencies, including externalities, game theory, and the tragedy of commons, revealing the complexities of individual decisions and their broader implications.

**Keywords:** Externalities, Prisoners' Dilemma, Tragedy of the Commons

### Introduction

Social science is concerned with how people make decisions and behave in group settings. Microeconomics is a social science that concerns itself with the interaction of individual and group behaviors in commercial settings. One such concern has to do with the supply and demand for goods and services in the economy. We will see that the microeconomic equilibrium of supply and demand is determined by various idiosyncratic interactions.

### Analysis

This article will discuss the structure of demand and supply in an economic market, the Prisoner's Dilemma, the impact of Externalities on society, and The Tragedy of The Commons. Furthermore, the article will explore how these concepts shed light on human behavior, decision-making processes, and the overarching societal implications, illuminating the complex interplay of individual and group behaviors in various social and economic contexts.

### The Microeconomic Concept of Supply and Demand

The microeconomic concept of supply and demand forms the cornerstone of market dynamics, influencing the allocation of goods and services in any economy. "Supply" refers to the quantity of a product or service that producers are willing to offer at different price levels. When the price of a good goes up, the supply increases because production becomes more profitable; when the price of a good goes down, the supply decreases. Supply is determined by the many costs that constitute production.

On the other hand, "demand" represents the quantity that consumers are ready to purchase at various price points. As the price of a good increases, demand decreases and vice versa. Demand is determined by the desirability of a good and the prices of competing goods. The quantity demanded and supplied for a particular good is subject to change. It is said that an economic market is in equilibrium when demand and supply are equal.

"Induced changes" in supply occur when external factors, such as technological advancements or alterations in production costs, lead producers to modify their output levels. Conversely, induced changes in demand result from shifts in consumer preferences, income levels, or changes in societal trends, which affect the quantity of goods or services that consumers are willing to buy. However, not all costs are included in the supply and demand equation.

Clearly, microeconomic phenomena exhibit a complex web of price, cost, and human behavior. Another notable phenomenon is externalities.

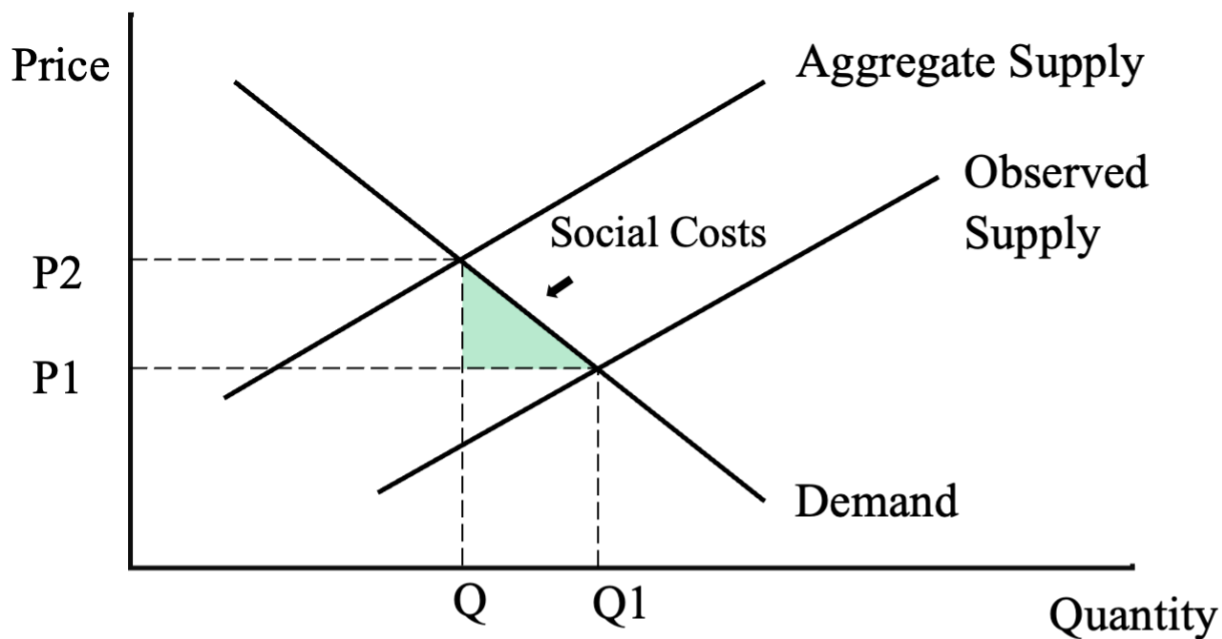
### Externalities

An "externality" refers to a situation in which the actions and decisions of one person or group have a consequential cost, usually negative but sometimes positive, on another person or group. When individuals or organizations partake in economic activities, they may not bear the full costs or consequences associated with their actions.<sup>[1]</sup>

Negative externalities can manifest in various forms, such as pollution, congestion, and noise. Air pollution from industrial factories is a negative externality. When factories emit pollutants, they harm the environment and public health, imposing costs on society. These costs include respiratory illnesses, ecosystem damage, and climate change. The factory owners may not fully bear these costs, leading to overproduction and insufficient pollution control. Negative externalities are illustrated in Graph 1 (below).

**Graph 1:** Negative Externalities





In the graph above, we see the demand and supply for a good, ignoring the costs of negative externalities. We thus derived a price of  $P_1$ , and a quantity supplied of  $Q_1$ . This market appears to be in equilibrium, but the negative externality has not been considered. If one includes the costs of externalities, expressing the higher costs including the externalities. We then derive new prices and quantities of  $P_2$  and  $Q_2$  respectively. The vertical distance between  $P_2$  and  $P_1$  represents the costs borne by society and not borne by the producer. The result is also that the quantity supplied is reduced.

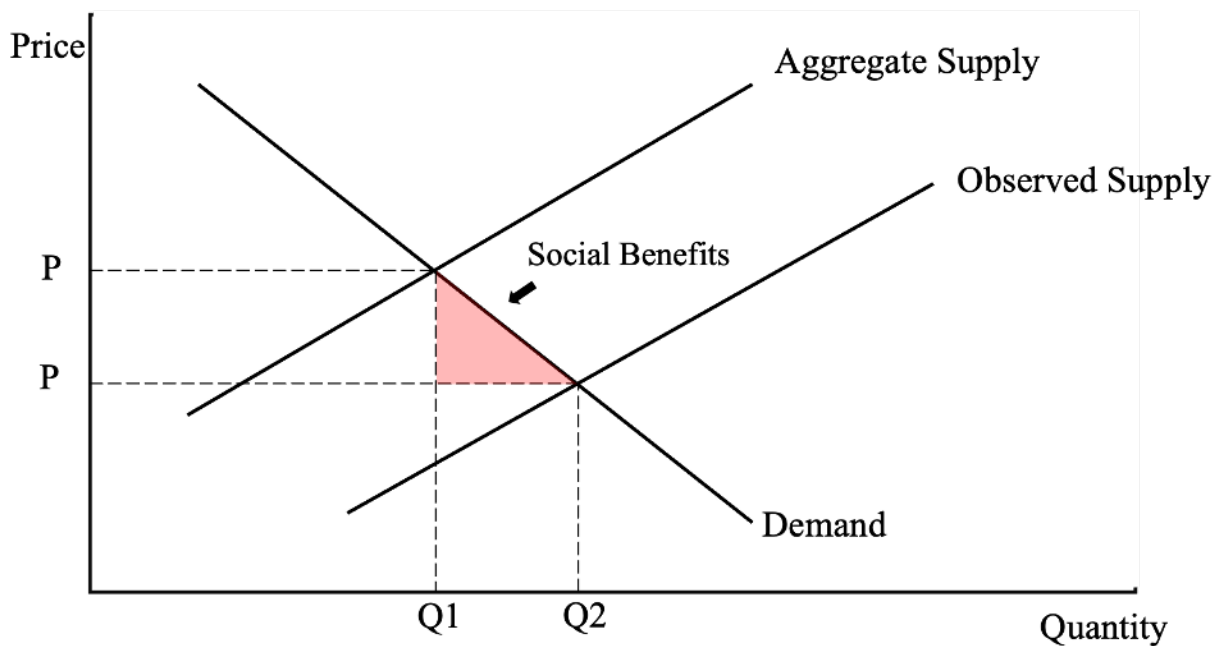
To address negative externalities, governments, and regulatory bodies often intervene through policies and regulations. They may impose taxes or levies on producers whose activities generate negative externalities, such as carbon taxes on greenhouse gas emissions. This payment shifts the social costs back to the producer.

Alternatively, a government may set emission standards or implement cap-and-trade systems to incentivize companies to reduce pollution. For example, there is a production limit or “cap” on one company. The company produced 500 units of pollution, and the limit is 1000 units. This company can trade at a price, the remaining 500 units to the other companies that produced more than the limit.

By recognizing the costs associated with negative externalities, these pollution taxes and similar measures aim to encourage responsible behavior and ensure that individuals and organizations consider the broader social and environmental impacts of their actions. In so doing, all costs are embodied in the supply and demand picture. Prices are thus higher, and supply is lower when all costs are observed.

A positive externality arises when an individual's economic behavior benefits others or society without imposing any costs on the beneficiaries. For example, your neighbor is planting flowers in his garden. As you passed by, you derived much pleasure from viewing the beautiful garden. The garden can give the community a better mood. The neighborhood becomes more valuable and real estate values increase due to the attractive garden. Graph 2 (below) illustrates positive externalities.

**Graph 2:** Positive Externalities



In the graph of positive externality, initially, the market seems to be in equilibrium, with a price of  $P_1$  and a quantity supplied of  $Q_1$ . However, this equilibrium does not reflect the presence of a positive externality. If economists incorporate the benefits of the externality, the observed supply must shift to the right, reflecting the additional benefits. Consequently, new equilibrium prices and quantities, denoted as  $P_2$  and  $Q_2$ , respectively, are derived. The vertical distance between  $P_2$  and  $P_1$  indicates the positive benefits that society gains, but which are not directly accounted for by the producer. Prices go down, and the quantity supplied goes up.

Positive externalities occur when an individual's actions generate benefits for others or society without receiving compensation for it. For instance, when someone installs solar panels on their roof, they not only reduce their electricity costs but also contribute to reducing overall carbon emissions and dependence on fossil fuels. The individual provides a free benefit to society. Positive externalities can arise in various sectors, including education, healthcare, and technology, where knowledge and innovations can “spill over” and benefit a larger population.

To encourage positive externalities, policymakers can implement measures such as subsidies or grants to incentivize individuals or organizations to engage in activities that generate positive spillover effects. While there is a social cost paid for by the government, the social benefit generated by an induced externality is presumed to be greater. If not, the government policy is faulty.

For example, governments may provide research grants to support scientific advancements that can be shared and utilized by society. By recognizing and rewarding positive externalities, societies can foster innovation, knowledge sharing, and collective progress.

Externalities, or spillover effects, were examined above at microeconomic market levels. This is a large picture economics concept. In the next section, we will examine how social interactions affect individuals in a smaller context. Both concepts represent social interactions at either an aggregate level or at a smaller local level.

### The Prisoner's Dilemma

The Prisoner's Dilemma arises in a hypothetical game unknowingly played by two prisoners. Here, the decisions made by each prisoner affect the other. The dilemma serves as a prominent example of a non-zero-sum game, illustrating that what may be individually optimal is not necessarily the best choice for both prisoners.

The Prisoner's Dilemma tells the story of two suspects who are caught by the police after committing a crime and detained in different rooms for interrogation. The police knew that the two were guilty but lacked sufficient evidence. The police impose upon the prisoners the following choices.

If both prisoners deny, each will be informed that he will be sentenced to one year in prison; if both confess, each will be sentenced to five years; if one confesses and the other denies, the one who confesses will be released, and the one who denies will be sentenced to ten years. Table 1 (below) illustrates the complexity of the Prisoner's Dilemma. The numbers in the table refer to years served in prison.

**Table 1:** The Prisoner's Dilemma

|           |         | Suspect B |        |
|-----------|---------|-----------|--------|
|           |         | Confess   | Deny   |
| Suspect A | Confess | 5 / 5     | 10 / 0 |
|           | Deny    | 0 / 10    | 1 / 1  |

The problem depicted by the Prisoner's Dilemma is that rationality can sometimes lead to group irrationality, that is, people may become trapped by their selfishness and thereby harm the interests of the group. It appears best for the individual to confess because he will be set free. If the accomplices trust each other, they will receive only one year. If they don't trust each other, the one who wants to get the maximum benefit will choose to deny, which will set him free. However, his accomplice will be sentenced to ten years. One's choice has a kind of spillover effect on the other.

Many people believe that the critical factor to the prisoner's dilemma lies in the lack of communication between the two prisoners. However, that premise can be challenged. For instance, proponents argue that the prisoners' dilemma arises from their inability to negotiate and communicate effectively. What if the prisoners were allowed to communicate?

If both prisoners trust each other, they will agree to deny and receive only one year in prison. However, if one of the two is untrustworthy, he will confess and go free, while the other will deny as agreed to receive the maximum sentence. The original assumption that communication is not allowed is not necessarily crucial to the outcome. In the presence of communication, it would be best for both to deny and to be sentenced to 1 year each.

However, this depends on the level of trust between the two accused. If one party chooses not to honor their agreement by denying, he will serve zero time, and his accomplice will serve ten years. Generally, criminals abide by a tacit code of conduct not to rat on one another without any communication. It is all a matter of trust.

The Prisoner's Dilemma itself emerges in many real-world contexts, such as instances of price competition, interpersonal relationships, and environmental conservation. In business, companies often face the dilemma of whether to cooperate or compete aggressively. In personal relationships, individuals grapple with trust issues and the temptation to prioritize personal gain over mutual benefit. Even in the realm of environmental conservation, nations may struggle with whether to cooperate in reducing emissions or pursue self-interest. This situation reveals the complexity of decision-making when conflicting interests are at play,

underscoring the need for strategic thinking and collaborative solutions in various aspects of human interaction.[2] Collaborative solutions in a spirit of trust generally provides the optimal aggregate outcome.

Another example of the damage that unregulated individuals or group behavior can impose on our resources is referred to as “the Tragedy of the Commons.”

### The Tragedy of the Commons

“The Tragedy of the Commons” highlights the inherent struggle between individual interests and the “common good.” In the public domain, there are economic resources that are shared by everyone. The term “commons” refers to any resource that is accessible to all members of a community, such as grazing land, forests, fisheries, and even water and air. A “Tragedy” occurs when individuals, acting independently and rationally, deplete or degrade a commonly held resource due to their pursuit of personal gain.

In this scenario, everyone seeks to maximize their benefit from the relevant resource. However, since no one has exclusive ownership or responsibility, there is a lack of incentive for anyone to conserve or sustain the resource. As a result, individuals tend to exploit the resource beyond its sustainable capacity, leading to its depletion or degradation.[3]

The Tragedy of the Commons underscores the limitations of unregulated self-interest and highlights the necessity of collective action and resource management.

In another example, imagine a group of herders facing an open pasture. Each herder wants to keep an extra cow because it is cost-effective – the revenue gained outweighs the cost of acquiring and maintaining the cow. However, as more cows are added, the average grass availability decreases, leading to lower per-unit output. Each herder may add one more cow, resulting in overgrazing. Unfortunately, this leads to insufficient food for all the cows, ultimately resulting, in the worst case, in starvation. Table 2 (below) illustrates this interdependency. The numbers in the 2 x 2 table (below) refer to “net incremental profits.”

As shown in Table 2, it becomes evident that the addition of a cow by one or both herders result in a troubling consequence: the aggregate output or profit diminishes to \$4. On the other hand, when both herders opt to maintain the current number of cows, their aggregate output or profit reaches \$6. This scenario serves as an illustration of the tragedy of the commons, emphasizing the importance of preventing resource depletion and striving for mutual gains among the herders. By collectively managing and conserving shared resources, they can avoid such detrimental consequences and ensure sustainable prosperity for all.

**Table 2:** The Example of Tragedy of the Commons Illustrated

|          |               | Herder 1  |           |               |
|----------|---------------|-----------|-----------|---------------|
|          |               | Add a Cow | No Change | Total Profits |
| Herder 2 | Add a Cow     | 4         | 5         | 9             |
|          | No Change     | 5         | 6         | 11            |
|          | Total Profits | 9         | 11        | 20            |

#### Summary:

Key concepts in microeconomic theory that were discussed included externalities, Game Theory, and the Tragedy of the Commons. These microeconomic concepts illuminated how individual actions can impact the wider society. Externalities are spillover effects of an individual's actions on others who are not directly involved in the transaction. Positive externalities confer benefits to the community beyond the individual. Negative externalities, such as pollution, inflict costs on society beyond the polluters' expenses.

The Tragedy of the Commons underscores the depletion of shared resources when individuals pursue their self-interest. Overgrazing in common pastures is a classic example, demonstrating how unchecked exploitation can result in resource degradation and inefficiency. These concepts underscore the need for governmental policies that eliminate externalities, promote cooperation, and prevent the tragedy that may emerge from shared resources, contributing to more efficient and sustainable economic outcomes.

Ideally, a spirit of trust and cooperation amongst all affected parties will optimize economic outcome for everyone.

[1] <https://www.mdpi.com/2071-1050/4/8/1776>

[2] <https://www.investopedia.com/terms/p...rs-dilemma.asp>

[3] <https://www.economicshelp.org/blog/g...externalities/>

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## 1.2: ESG Performances and Economic Growth

**Title:** ESG Performances and Economic Growth

**Author:** Yifei Zhu

**Keywords:** CEO compensation, ESG, governance, disclosure, stock liquidity, risks, county-to-district (CtD) reform, economic growth, correlation

**Abstract:**

This paper investigates whether better ESG performance enhances economic growth. The study sample covers both the MSCI universal ESG performances of the USA, China, Europe, and Japan, and their annual GDP growth rates from 2010 to 2023. The results show that ESG performance has a positive relationship with economic growth. This relationship is much stronger in developed countries' samples. Moreover, developed countries have more positive ESG correlations with developed countries than with developing countries.

### Introduction

ESG refers to “Environmental, Social, and Governance.” Environmental refers to the sustainability of production and the degree of environmental friendliness. Social stands for the responsibility to employees, such as the implementation of gender equality, and the firm’s impact on community and society. Governance speaks to the control mechanism over ownership and management, including transparency, bribery, and internal control.

In the next section, we review how leadership, disclosure, risk, and other factors influence ESG performance. “ESG performance” refers to a firm’s level of responsibility to the three ESG factors. The better the ESG performance is, the less risky the company would appear to investors. What’s more, ESG disclosure may have a positive effect on operations, financial and marketing performance, which improve firms’ ESG.

### Literature Review

Alareeni et al (2020) measured ESG performance by incorporating numerous variables as noted in Table 1 below. Some of the variables include financial measures, such as return on assets, return on equity, Tobin’s Q ratio, among others. Other variables refer to the company’s ESG disclosure.

**Table 1:** Description of Variables

| Variable | Definition                                                  | Description                                                                                                                                                                                                                                                                                            |
|----------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Perf     | Firm's performance                                          | <ul style="list-style-type: none"><li>• Measured by ROA, ROE and Tobin's Q.</li><li>• ROA equals net income divided by total assets.</li><li>• ROE equals net income divided by equity.</li><li>• Tobin's Q equals current liabilities plus the market value of share capital divided by TA.</li></ul> |
| ESG      | Firm disclosure of its environmental, social and governance | <ul style="list-style-type: none"><li>• The ESG Bloomberg index includes all the disclosure of EVN, CSR, CG indices.</li></ul>                                                                                                                                                                         |
| EVN      | Environmental disclosure                                    | <ul style="list-style-type: none"><li>• The EVN Bloomberg index measures the disclosure of energy use, waste, pollution, natural resource conservation and animal treatment.</li></ul>                                                                                                                 |

|     |                                  |                                                                                                                                                                                  |
|-----|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CSR | Social responsibility disclosure | <ul style="list-style-type: none"> <li>The CSR Bloomberg index measures disclosure of business relationships, donation, volunteer work, employees' health and safety.</li> </ul> |
| CG  | Governance disclosure            | <ul style="list-style-type: none"> <li>The CG Bloomberg index measures the disclosure of corporate governance code.</li> </ul>                                                   |
| FS  | Firm size                        | <ul style="list-style-type: none"> <li>Measured by total assets.</li> </ul>                                                                                                      |
| FL  | Financial leverage               | <ul style="list-style-type: none"> <li>Measured by total debt to total assets.</li> </ul>                                                                                        |
| AT  | Asset turnover                   | <ul style="list-style-type: none"> <li>Measured by net sales as a percentage of total assets.</li> </ul>                                                                         |
| AG  | Assets growth                    | <ul style="list-style-type: none"> <li>Measured by the annual change in total assets.</li> </ul>                                                                                 |

ESG disclosure has great impact on individuals (Alareeni and Hamdan, 2020). Knowing this, firms will exploit ESG in order to successfully market products to individuals thereby increasing product sales and corporate profits. Greater sales and operating profits will enhance corporations' abilities to service their debts thus reducing credit risks. Additionally, environmentally sensitive corporations will attract greater shareholder investment. Based on the research, it is suggested that national governments and supranational policymakers, such as the World Bank, should require firms to accurately disclose their ESG activities to enhance financial markets stability worldwide. They also suggested the following results:

The descriptive analysis results show that the overall ESG, CSR, EVN and CG tend to be higher with firms that have high assets and high financial leverage. Further, results showed that firms with high disclosure levels of ESG, EVN and CSR have higher operational and financial performance (ROA and ROE). (Alareeni and Hamdan, 2020)

A firm's ESG performance is influenced by its nationality, its boards of directors, and its ownership (Gillan et al, 2020). First, nationality can be further broken down into a country's legal system, its economic development and education. A country's legal system has the greatest impact among these factors.

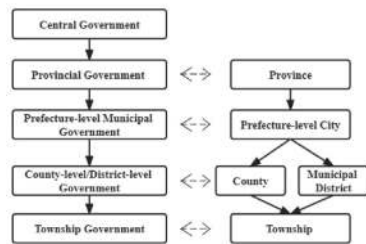
What's more, ESG would be higher if the firm had either a female or a young chairman in the board. Family ownership satisfies investors' ESG expectations according to Gillan, Sekerci, and Starks (2020), but it is contradicted to the results that family ownership has lower ESG scores due to the expropriation hypothesis of family ownership by El Ghouli, Guedhami, Wang, and Kwok (2016). This difference might be caused by different country-based samples. Besides, state-owned firms usually perform better as state-owned firms have better ESG score, including aggregate, Environment and Social performances.

ESG performance has both positive and negative effects on the firm (Gillan et al, 2020). ESG attracts more investors who believe that ESG will bring positive long-term performances. Young people usually have more incentive to invest in firms with better ESG performance. It is suggested that there is a positive relationship between reputation of ESG and following implicit contracts, but some researches had contradict results. However, higher ESG performance will reduce ROA or stock returns as people assume that "any benefits to stakeholders from social responsibility come at the direct expense of firm value."

The influences of government policies for ESG performances can be explained by the following research. Xue (2023) found that rent-seeking behavior, which occurs when entities gain wealth without making any contribution to society, affects ESG performance. The result shows that county-to-district (CtD) reform can solve the problem of rent-seeking behaviors of businesses. Figure 1 below explains China's government hierarchy and administrative division. The left column represents the government hierarchy, and the right column represents corresponding administrative division. (Xue et al., 2023, p. 4)

By weakening the autonomy of the county-level government and improving integration among different districts, which enhanced firms' competitiveness. The firms may therefore pay more attention to ESG strategies. Local governments would also make more pro-ESG policies while having more control over the firms. The reform improved not only the regulation in the district on rent-seeking behaviors, but the ESG performance of the firms.

**Fig. 1** China's government hierarchy and administrative division.



Deng and Cheng (2019) suggested that the government should make pro-ESG policies. It was suggested that the government should focus more on manufacturing, and consider the differences in the volume of pollution and products among small firms. For example, the government should provide more subsidies and more environmental protection investment to firms that produce more pollution. Entrepreneurs and investors should also be encouraged to manifest higher social responsibility by developing or investing in industries with higher ESG evaluation.

Based on this, Xue believes that centralization of prefecture-level municipal governments can improve the ESG performances of the companies. Xue suggested that the government can weaken market segmentation, ease economic uncertainty and reduce rent-seeking by using ESG.

Deng and Cheng (2019) suggested that policies should be made based on different features of firms and focus on switching secondary industry to a more environmentally friendly one. Investors and entrepreneurs should also focus more on firms with better ESG performances.

ESG disclosure can reduce information asymmetry. The effects of firms' ESG performances on its finances would be more visible to investors. This will increase the accuracy of companies' stock pricing and thereby reflect true intrinsic values. Thus, investors will have more confidence investing in stocks that engage in ESG initiatives. (Sassen et al, 2016)

Higher social performance can also reduce financial risk by increasing firms' flexibility when facing economic downturns. This also enhances firm reputation by being a good member of the community and respecting business ethics. (Sassen et al, 2016)

Premium pricing due to Environmental and Social activities have a unidirectional effect on firm risk. This means that Environmental and Social premiums can affect firm risk while firm risk cannot affect the premiums. (Sassen et al, 2016)

Chen et al (2022) found that ESG performances have a positive and profitable relationship with the Chinese Sovereign Wealth Fund, which in turn attracts more investment. It was also found that among the three factors of ESG, the governance factor has a stronger correlation with the amount of additional investment attracted than the other two.

Overall, as Gillan et al (2020) describe, local governments' policies would influence firms' ESG performances very significantly. Features of a firm, like boards and ownership, would also affect its ESG performance. Thus, a firm with better location and more equal board structure will have better ESG performance.

Alareeni and Hamdan (2020) state that ESG disclosure can improve a firm's asset efficiency and firms' assets market value by improving the operational and market performances. ESG can be used as a strategic product for firms to increase profit.

According to Xue (2023), government policies such as CtD reform can reduce rent-seeking behavior by which companies seek to derive economic benefits without having made any ESG contributions. The CtD reform combined small districts and led to more unified policies, which reduced information asymmetry between investors and firms, so that investors are able to invest for the long run and thereby foster greater market stability.

Sassen et al (2016) found that a firm with higher ESG value will have lower risks, including systematic, credit and total risks, and have a better reputation, especially for environmental industries. Policy makers should make regulations for compulsory ESG disclosure which would assure the accuracy of the data and stabilize the stock price of corporations. They also suggested that:

To provide (potential) investors with relevant and reliable information, national or supranational policymakers could increase disclosure regulation (e.g., by means of mandatory CSR disclosure). Moreover, mandatory disclosure could encourage firms to integrate ESG factors into their firm strategy and operations and thus stabilize capital markets by lowering market fluctuations. (Sassen et al, 2016)

Chen et al (2022) found that attention to ESG factors could attract investors. Chen also found that the governance factor has a stronger impact on attracting investors than environmental and social factors.



In summary, locating in a country with an established and viable legal system, positive social attitudes toward ESG implementation, equal leadership chances for every member in the firm and community, and transparent disclosure can significantly improve a firm's ESG performance and reduce its financial risks. Government policies and reforms are also very significant. Policies that encourage disclosure and aim for a more equal and competitive market help to promote ESG.

### Analysis

ESG is being implemented in order to create a more sustainable production infrastructure and thereby to attract more equity in investors. An unintended consequence is that ESG performance contributes to economic growth as measured by GDP. Improving Environmental performance can help to attract more investment, especially from younger investors. Having better Social performance means having more equal working opportunities for women and vulnerable groups. Thus, more tax revenue may be created as more people are going to work. Additionally, enhancing Governance performance can improve firms' performance and thus improve investment and directly improve economic growth.

As calculated in Table 2, real GDP and GDP per capita are highly correlated. This is because the population and productivity have not changed much through the years. Thus, it appears that productivity is not affected by ESG performance. ESG performance can increase investment and thereby enhance economic growth while not enhancing productivity.

**Table 2:** International Correlations of GDP and GDP Per Capita from 2009 - 2023

|                         | <u>U.S.A.</u> | <u>EU</u> | <u>China</u> | <u>Japan</u> |
|-------------------------|---------------|-----------|--------------|--------------|
| Correlation Coefficient | 0.9996        | 0.9972    | 0.9999       | 0.9947       |

According to the World Bank (1996), a country's stock market has a positive effect on the country's economic growth, as stock market liquidity is positively and significantly correlated with GDP growth. To show ESG performance, we chose the MSCI ESG universal annual index. Table 2 (below) provides data for MSCI ESG universal annual performances for four countries. A positive trend of improving "ESG Profiles" means that there is an increasing tendency for firms to improve ESG. This is measured by the percentage increase of ESG performance over the years provided. The table depicts the performance of companies with identifiable ESG characteristics and positive trends. These indices can be used to represent the various countries' ESG changes, because the indices comprehensively include large and mid-cap companies. The table shows that there were decreases in all countries' respective performances in 2018 and 2022.

**Table 2:** MSCI ESG Universal Annual Performance (%)

| <u>Year</u> | <u>USA</u> | <u>EU</u> | <u>Chn.</u> | <u>Japan</u> |
|-------------|------------|-----------|-------------|--------------|
| 2010        | 12.81      | 4.5       |             |              |
| 2011        | 0.62       | -9.9      |             |              |
| 2012        | 13.28      | 19.42     |             |              |
| 2013        | 33.27      | 26.11     | 8.6         | 54.11        |
| 2014        | 13.25      | -5.37     | 9.18        | 10.19        |
| 2015        | 0.67       | -0.88     | -6.93       | 10.6         |
| 2016        | 10.94      | -1.7      | 2.17        | -0.62        |
| 2017        | 21.7       | 25.84     | 55.33       | 19.2         |
| 2018        | -4.23      | -14.84    | -19.44      | -15.15       |
| 2019        | 31.9       | 25.53     | 26.34       | 19.85        |
| 2020        | 21.51      | 8.64      | 30.34       | 9.24         |
| 2021        | 27.5       | 16.84     | -20.08      | 13.01        |
| 2022        | -19.92     | -16.29    | -22.6       | -4.64        |

|      |       |       |        |       |
|------|-------|-------|--------|-------|
| 2023 | 25.76 | 20.39 | -12.06 | 27.92 |
|------|-------|-------|--------|-------|

The declines in 2018 and 2022 might be explained by the U.S.A.'s withdrawal from the Paris Agreement in 2017 and the Omicron epidemic which started toward the end of 2019. The Paris Agreement is an international agreement, aiming to reduce pollution and protect the environment. Nearly 200 countries are signatories to the agreement, including the European Union as one country. The U.S.A.'s withdrawal from the Paris Agreement led to less overall financial support of environmental protection, causing a general fall in ESG performances. This might be proved by halted climate finance after the withdrawal. According to the Times (2025), contributions promised by the Obama administration to the Green Climate Fund (GCF) were canceled. Only a small part of the \$3 billion commitment was honored. This led to a decline in ESG. To explain the decline in 2022, it is possible that the epidemic had a lagging negative effect on labor activity which is a major component of ESG performance.

Table 3 (below) shows the annual GDP growth rate of the countries studied. In most of the years, the countries manifest positive GDP growth rates. In 2020, however, most countries manifested negative growth; there was a slight increase in growth in 2023.

**Table 3:** Annual GDP Growth Rates (%)

| Year | U.S.A | European Union | China | Japan |
|------|-------|----------------|-------|-------|
| 2010 | 2.7   | 2.23           | 10.64 | 4.1   |
| 2011 | 1.56  | 1.9            | 9.55  | 0.02  |
| 2012 | 2.29  | -0.7           | 7.86  | 1.37  |
| 2013 | 2.12  | -0.08          | 7.77  | 2.01  |
| 2014 | 2.52  | 1.6            | 7.43  | 0.3   |
| 2015 | 2.95  | 2.31           | 7.04  | 1.56  |
| 2016 | 1.82  | 1.97           | 6.85  | 0.75  |
| 2017 | 2.46  | 2.84           | 6.95  | 1.68  |
| 2018 | 2.97  | 2.07           | 6.75  | 0.64  |
| 2019 | 2.47  | 1.81           | 5.95  | -0.4  |
| 2020 | -2.21 | -5.65          | 2.24  | -4.15 |
| 2021 | 5.8   | 6.01           | 8.45  | 2.56  |
| 2022 | 1.94  | 3.48           | 2.99  | 0.95  |
| 2023 | 2.54  | 0.45           | 5.2   | 1.92  |

We have posited that ESG implementation positively impacts economic growth and in turn ultimately produces stock market gains. This paper correlated ESG implementation with economic growth, however no identifiable lag could be determined. Given the data in Tables 2 and 3, it appears that there is a lag of approximately two years between ESG implementation and the manifestation of economic growth.

Table 4 (below) presents the correlation matrix of ESG performance and GDP growth. ESG performance has a strong and positive relationship with economic growth, given a two-year lag.

**Table 4:** Correlations of ESG Performance and GDP Growth (Two Years Lag)

| <u>U.S.A</u> | <u>EU</u> | <u>China</u> | <u>Japan</u> |
|--------------|-----------|--------------|--------------|
| 0.66         | 0.591     | 0.216        | 0.594        |

Table 5 (below) shows the correlation coefficients of each country's relative ESG performance. It shows that there is a close relationship between the U.S.A. and Europe, and between Europe and Japan, but not between China and Japan.

**Table 5:** Correlation Matrix of Relative Countries' ESG Performances

|        | <u>U.S.A.</u> | <u>Europe</u> | <u>China</u> | <u>Japan</u> |
|--------|---------------|---------------|--------------|--------------|
| U.S.A. | 1.00          | 0.89          | 0.49         | 0.75         |
| Europe | 0.89          | 1.00          | 0.55         | 0.82         |
| China  | 0.49          | 0.55          | 1.00         | 0.33         |
| Japan  | 0.75          | 0.82          | 0.33         | 1.00         |

Improvement in ESG appears to have a positive effect on future economic growth, given a two-year lag. The positive effect is stronger in developed countries, such as the USA, than in developing countries, such as China. This might be explained by investors' preferences in developed countries to be more environmentally friendly, better structured, and more socially beneficial means of production.

Additionally, there is a stronger positive correlation between developed countries, such as the U.S.A and Europe, than between developed and developing countries, such as China and the U.S.A.. However, as the ESG indexes cover just one decade, more data and years may present different findings. Moreover, causality between ESG and economic growth cannot be established.

### Conclusion

There is a general growth in both GDP and ESG performances in the four countries studied from 2010 to 2023. It was found that improved ESG performance positively affects national economic growth. The correlation between ESG performance and GDP is stronger in developed countries than in developing countries. This might be caused by more attention to ESG in developed countries. Furthermore, ESG performances also are more strongly correlated among developed countries than among developed and developing countries. There is no evidence that ESG enhances productivity.

### Suggestions for future research

As time passes, more ESG data should become available. Future research may focus on the extent of the time lag between ESG and economic growth. Future researchers ought to consider the impacts of confounding variables on economic growth and stock market performance.

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## 1.3: A Proposal for the Sustainability of Peking Opera

**Title:** A Proposal for the Sustainability of Peking Opera

**Authors:** Yueqi Cynthia Li

**Keywords:** Peking Opera, Market, Decline, Culture

### Abstract:

Peking Opera, a significant part of Chinese cultural heritage, is declining. The challenges it faces include an aging audience, competition from modern digital entertainment, and high-ticket prices. The paper applying marketing concepts “4 Ps” (Product, Price, Place, Promotion) and “5 Cs” (Customer, Company, Collaborators, Competitors, Context) to help the opera regain popularity in modern entertainment market. Solutions include co-branding with popular brands, adjusting ticket prices for broader accessibility, and promoting the art through social media and digital platforms. Additionally, the paper emphasizes the importance of engaging fan culture and international collaborations to boost Peking Opera’s global presence. These efforts shorten the gap between tradition and modernity, help traditional opera become popular.

### Introduction

Peking Opera, one of the most representative arts of China, was known as “the quintessence of Chinese culture”. Originating from the Qing dynasty, in 1790, Peking Opera absorbed techniques and methods various local operas, notably, the K’un Chu and the Hwei Ban. Peking Opera is divided into traditional opera and modern opera. Each play showcases a wonderful story in real life.

Peking Opera is composed of four typical role types: Sheng, Tan, Jing, and the court jester. Different roles have different characteristics, which represent a specific group of people (Liu).

Music is a dispensable part of Opera. The three main instruments are the Hu Chin, Erh Hu, and the Big Drum. “The Orchestra is to accompany all the movement of every actor or actress on the stage and the drummer is the conductor of the orchestra.” (Liu)

From a hundred years ago, Peking Opera was open to all the public, across ages and social classes. However, Peking Opera faces challenges from different dimensions now. Gen Z in China barely knows about such art format and the original audience group has aged. Externally, with the development of technology, many allures distract the young generation. They are more willing to spend time on computer games, social media, and shopping. Those relatively low-cost entertainments are easier to enjoy. As people have already adapted to the fast-paced life, they lost the patience to sit down and watch this “old-fashioned” performance. Thus, few people would like to be fully devoted to Peking Opera.

Internally, Peking Opera has deficiencies in performance form. The long and slow tone bores the audience easily. The special pitch is hard for some people to accept. Each normal play is at least one and a half an hour, so it is a torment for people who don’t have any background knowledge of this art.

### History

Peking Opera has had a high status in Chinese society for centuries. As a traditional art that originated from the 18th century, it was supported by the government and normal people. However, it cannot follow the change of contemporary society, which is adopted by technology and fast-paced daily life.

The ticket price for Peking Opera is also not affordable to most of the audience, so fewer people walk into the theater for this art. It seemed to be doomed to fade out of the marketplace. It was unable to compete with the popularization of film, television, and diverse forms of pop culture (Zhang).

### Literature Review

Previous studies show that the government has realized this problem. Several related policies have been enacted. Mass Entrepreneurship and Mass Innovation (MEMI) was established, promoting an entrepreneurial solutionism to exploit market values and to speed up China’s long-term goal of developing its economy through technological innovations (Li).

Not only does Peking Opera face the problem of declining affordability, but also the film industry. Mingxing Company brought the film industry to Chinese people’s sights in the 20th century. This company was not established with major investment by powerful industrialists or capitalists but with a small fund raised by its founders and prospective shareholders (Huang 23-68).

Fan culture can bring indispensable income to traditional culture. As audience participation has long been an indispensable component in the development of traditional opera, the contextualized fan culture is not a completely novel form, but an extension of the traditional mode situated in the age of information. “The decrease of spatiotemporal restrictions and economic costs would help traditional opera become popular again.” (Zhang)

People who were employed in culture enjoy consuming Chinese art and culture. Artists possess more desire than normal audiences to learn Chinese Opera. Thus, NACTA (National Academy of Chinese Theatre Arts) should invite more foreign artist coming to China to learn it and then spread it to the world. (Link)

Some research has been done to find an effective way to spread traditional culture to the world. Relative data analyzing the action of introducing Chinese Opera to foreign artists and normal audiences proves that foreign artists can spread it quickly. This is a more efficient way to popularize Peking Opera globally.

The investment in the diversity of inheritance ways should be strengthened. The government should provide support in system and policy; the Education institution of Peking Opera should have its own characteristic, in order to improve teaching efficiency; sorting out and filing work should be done (Zhu).

According to the data from a ten-year experiment, Chinese Opera is high art, so it's hard to be interpreted by normal audiences, even artists who focus on other arts. Artists possess more desire than normal audiences to learn Chinese Opera. People who were employed in culture like to consume Chinese art and culture (Lin).

## Analysis

Peking Opera is worth saving. It contains the essence of Chinese culture. The story of each show is about Chinese traditional habits and popular stories. It tells audiences meaningful principles. The instrument Opera is special, it contains a hundred years of musicians' wisdom. It is not too late to overcome the temporary difficulties it faces.

Different from most of the previous studies that focus on solving the problems Peking Opera faces through culture studies, this paper is going to discuss them from the aspects of marketing and consulting. The “4 Ps” marketing strategy, proposed by E. Jerome McCarthy, is the basic logic of solving a marketing dilemma. It's the abbreviation of the four key elements in marketing strategy: Product, Price, Place, and Promotion (McCarthy).

To cater to the entertainment market, Peking Opera, as a product that audiences should pay for, should change itself to a more attractive product. Co-branding is a good option. For example, co-branding with Hey Tea, a popular Boba Tea brand in China, can help young people understand traditional culture when enjoying the drink. The Opera elements are designed on the packing of the cups. If people get boba tea, they will notice the elaborate decoration and may get curious about this culture.

The opera company should divide the ticket prices into different levels. Thus, people with less consumption power can also watch films by buying low-price tickets. This measure helped film art spread quickly. The price of one ticket for a Peking Opera performance is at least 30 dollars now. Many people cannot afford it. The method of Mingxing Company can be adopted.

Beijing, as the origin of Peking Opera, should be the main spot for popularizing this art. The peripheral products should not only be sold at the stores open in the theatre but also sold in normal markets. This helps Peking Opera break the stereotype of “expensive” and “extravagant”.

Now, the ways of promoting Peking Opera are too boring and rigid. The government suggests posting posters on the walls beside the streets. However, the design of those posters is too simple and fails to attract pedestrians. To increase popularity, Peking Opera can open online stores to promote products. Furthermore, adapting the opera singing aria and combining it with pop music is an efficient way to popularize this art.

Fan culture brings commercial value to Peking Opera, driving the revenue for this art. As the main platform of fan culture, the internet is an indispensable way to spread this traditional art to more people around the world. With the development of TikTok, Instagram, YouTube, etc., popular internet celebrities' salaries are increasing. The income from operating the Peking Opera business is increasing. To be more competitive in the entertainment market, the Opera must be more proactive in advertising on social media and selling proprietary products including opera dolls, bookmarks with opera elements, and desserts that are carved as opera faces.

The five C market strategies focus on fulfilling customers' requirements. “The central placement of target customers reflects their defining role in the market; the other three entities--the company, its collaborators, and its competitors--aim to create value for

these customers.” As customers’ preferences transfer to social media nowadays, the products of Peking Opera should follow this trend of popularizing on social media.

The competitors of traditional opera are what the young generation is addicted to nowadays, computer games, pop music, and social media, for example. Directly competing with them is impossible. Peking Opera should be combined with these modern elements. With these collaborations, Peking Opera can be recognized by more people. Thus, it can have a position in the entertainment market.

Furthermore, to spread Peking Opera globally, relative institutions in China should invite international artists. Watching this art in person helps them feel the deep connotation of it. Thus, NACTA (National Academy of Chinese Theatre Arts) should invite more foreign artist coming to China to learn it and then spread it to the world.

## Conclusion

The Peking Opera faces several challenges during this modern and fast-paced era. Such decline is mainly attributed to the audience demographics shift and the booming of other digital entertainment, like social media and computer games. To help the opera overcome the dilemma, strategic adaptation is essential. The use of marketing strategies such as the “4Ps” and “5Cs” theories builds the bridge for Peking Opera between tradition and modernity. The approaches, like co-branding with popular consumer brands, adjusting ticket prices, varying distribution channels, and being active on social media, can decrease the gap with younger generations. Additionally, utilizing modern elements and combining them with fan culture and digital marketing strategies, can amplify accessibility. Moreover, collaborating with international artists and institutions can spread the notability of Peking Opera globally.

These marketing and consulting strategies help Peking Opera regain its place in the entertainment market. Embracing these changes will not only revitalize Peking Opera but also reinforce its position as a beacon of Chinese cultural heritage for generations to come.

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## 1.4: Market Failure in the Food Industry in North America

**Title:** Market Failure in the Food Industry in North America

**Authors:** Junle (Jula) He

**Abstract:**

This paper identifies all the major causes of market failure in the food industry in North America. We present a cross-regional comparison, citing the ways in which these failures manifest differently in the US, Canada, and Mexico given their different regulatory environments, market structures, and consumer behaviors. Additionally, we note that large-scale agriculture, with its concentration of market power, economic inequality, and numerous marketing practices in combination exacerbate underlying market failures. We find that policy solutions require a holistic approach, including stricter regulatory frameworks, increased public investment, market-based incentives, and public awareness campaigns with a focus on social equity.

**Keywords:** market failure, food insecurity, monopoly, market concentration, externality, imperfect information, public good

### Introduction

This paper provides a comprehensive exploration of market failures within the food industry, drawing from a meta-analysis of existing literature to offer a summary and update on current perspectives.

The food industry in North America includes various sectors from agricultural and food production through processing and distribution to retail and food services. The food industry alone contributes 1.4% (Government of Canada) and 5.6% (US Department of Agriculture) of Canada and the United States' GDP in 2023 respectively. It also employs millions of people and shapes the eating patterns of hundreds of millions more. Despite its importance, the food industry is facing large challenges that have resulted in market failures across the industry, harming efficiency and equity in markets.

Market failure is when the free market pursuing private interest is unable to operate at the socially optimum level. There is an inefficient use of society's resources. The market clearing forces do not maximize social surplus. The inefficiency could be caused by externalities, public goods, asymmetry of information, or market power. Food has many underlying factors that create market failure related to the environment, human health, food security, and more that will be discussed in this paper. This paper will provide a detailed examination of the various market failures, their causes, consequences, and possible solutions to ensure that the food industry is in a better position to serve society's needs.

### Terms

- **Market failure** occurs when the free market, driven by private interests, fails to operate at a socially optimal level, leading to an inefficient allocation of society's resources and a failure to maximize social surplus.
- **Food insecurity** is broadly defined as a condition where people in affected households struggle to provide enough food for all members due to a lack of resources, as noted by Elmes (2016). To achieve food security, Rocha (2007) emphasizes that several criteria must be met: availability (sufficient food to meet people's needs), accessibility (assured physical and economic access to food), adequacy (nutritious and safe food that supports healthy lifestyles and is produced sustainably), and acceptability (food that is culturally appropriate and obtained in ways that uphold dignity, self-respect, and human rights).
- **Economic inequality** is the unequal distribution of resource allocations, access to productive assets, and compensation for labor within a society restricts individuals' ability to meet basic human needs.

### Literature Review

This literature review explores the pervasive market failures within the food industry, examining how unregulated markets often fail to address critical issues such as food insecurity, environmental degradation, and public health. It synthesizes key insights from scholars like Rocha (2007), who identifies negative externalities, imperfect information, public goods, and market power as fundamental sources of market inefficiencies. Further contributions from Lusk (2013), Steier (2011), Elmes (2016), Collart and Interis (2018), and Marcone et al. (2020) expand on these themes, investigating the impact of industrial agriculture, economic inequality, advertising, and consumer behavior on food systems. Rocha (2007) examines how the unregulated market fails to address food insecurity and proposes interventionist policies. She identifies four sources of market failure in the food system.



## Negative Externalities

Rocha (2007) highlights environmental externality as a significant source of market failure, with an example being the **pollution** generated by ‘**factory farms.**’ A factory farm is a form of intensive agriculture designed to maximize profits using as few resources as possible; it’s a method of raising farmed animals usually in indoor ‘factories.’

The disposal of waste generated by factory farms in the process of producing, transporting, or consuming a given commodity frequently causes pollution. In the operation of factory farms, they pollute the ground water which increases the cost of purifying water for communities nearby. Factory farms also cause air pollution through the emission of chemicals like hydrogen sulfide and ammonia, contributing to global warming. Pollution leads to adverse health effects, increasing the risk of diseases for individuals. With these health effects of pollution, the external costs of factory farms also include sickness which can contribute to missed work and even death in extreme cases. Similarly, pollutants can be toxic to sensitive plants and trees, hindering the growth of natural species. Lastly, factory farms also produce a foul smell which reduces the value of neighboring properties.

These pollution costs are called negative externalities because they impose costs on society in the course of the production process which are not paid for by the producer nor the consumer. Though consumers may seem to be paying a lower price for the meat produced at factory farms, the external cost of food production is imposed on society at large and is not reflected in the price of the good. This shows the inefficiency of factory farms from a social perspective.

The modern agricultural industry also produces other externalities. For example, the highly intensive plowing and cropping leads to soil degradation and erosion. Accordingly, soil becomes less productive for further food production which decreases crop productivity and groundwater quality. The expansion of agricultural activities and use of chemical pesticides and herbicides have also contributed to deforestation. This loss of biodiversity means a loss of ‘carbon sinks’ to combat air pollution and climate change as well as a loss of local foods and local food habits. Rocha proposes several policies that could help combat pollution negative externalities. One solution is to impose high pollution taxes. The tax would increase producers’ private cost of production which can act as a disincentive to pollute. Similarly, the government can regulate operations by setting a maximum limit on the number of animals per farm which would reduce the average level of pollution. The government could also set rules on what type of waste treatment must be used to prevent pollution.

Another example of a negative externality that Rocha references relates to the overexploitation of common access environmental resources. These resources are considered “non-excludable” and “rivalrous.” Non-excludable means that it is impossible to exclude anyone from sharing in the use or consumption of the resource. Rivalrous signifies that the consumption of the resource by one person limits the amount left for other people.

Fish in the ocean are a very commonly overexploited resource. Though fish can reproduce themselves, they always tend to be harvested at a faster rate than they can naturally replenish themselves. This is because each fisherman has the incentive to take as much fish as possible and as fast as possible to earn more money. No fisher intends to preserve present stock because their sacrifice will not pay off if other ‘free riders’ do not do the same. Since everyone will fish without control to act in their self-interest, this leads to the depletion of the renewable resource of fish, leaving no more for future generations and leading to the tragedy of the commons.

An effective way to address the problem of common access resource of fisheries is to set quotas on the amount that can be fished. However, setting an appropriate quota level may require high amounts of data collection which adds to the cost of the policy. It also requires constant monitoring and regulation enforcement which makes setting quotas unfavorable in some cases. Alternatively, there could be government requirements such as licenses that would make it less accessible and harder to fish, helping mitigate the depletion of the resource. Ecologists like Garrett Hardin have also argued for private property as a solution to control overexploitation.

The most favorable solution is the creation of ‘developing markets.’ This is essentially the creation of market mechanisms for common property resources. By assigning economic value to environmental services, these markets incentivize sustainable resource management and conservation. They create economic opportunities for local communities, especially in developing regions, by tapping into new revenue streams. For example, in fishing, the government can assign specific portions of the total allowable catch (TAC) to individuals or companies. These quotas can be bought, sold, or leased, creating a market for fishing rights. In addition, fishermen can receive payments for engaging in practices that benefit the ecosystem, such as reducing bycatch or protecting breeding grounds.

## Imperfect Information

Imperfect information is another common source of market failure in the food industry. Taking the same example of agriculture, some farmers do not realize that they can prevent soil erosion by implementing better farming practices, like crop rotation and reduced tillage. However, in most cases, farmers are aware and have information to prevent soil erosion. In fact, they are merely limited by their economic condition. Many farmers who live in poverty depend on agricultural products they produce on the day for the survival of their family. They will proceed with their current producing conditions at the cost of lower food productivity because they don't have sufficient income to invest in new techniques. In addition, the land rights of their farms are often uncertain or inexistent. Therefore, farmers have no incentive to invest in soil preserving techniques on a land that they do not own or that might be taken away from them anytime.

To resolve the root of the issue, Rocha suggests that the best strategy is to provide rural credit for adopting soil preservation techniques. Rural credit refers to the provision of loans and other financial services to farmers and rural communities which can help farmers meet their financial needs to implement new techniques. This will help farmers improve their productivity in the long run, manage risks, and ensure the sustainability of their agricultural practices. Rocha also mentions that securing land rights can assist small farmers in making informed decisions about the environmental impact of their practices and maintaining the long-term productivity of their lands for future food production. This can include land titling and registration programs that would ensure farmers' legal ownership or long-term use rights.

## Public Goods

Market failure can also arise when private markets fail to supply sufficient quantities of an activity or good, even when its benefits significantly exceed their social costs (as seen with positive externalities). This is the case for public goods characterized by the features of non-rivalry and non-excludability. Being non-rivalrous means that the good can be enjoyed by many people without taking away the amount that can be consumed by others. On the other hand, non-excludable means that it is impossible to prevent people from using them, even those who have not paid for them. Free markets often underprovide or do not provide public goods at all because there is no profit motivation to lead private firms to supply a socially efficient quantity of such goods. Therefore, individuals will act in their self-interest despite the fact that public goods generate high benefits to society.

Food security is seen as a public good; environmental quality and food safety are obvious public goods. Nutritious foods and healthy diets, though they can be private goods because they are rival and exclusive, have wider impacts. When these are insufficient, it leads to public health issues, like higher social and economic costs due to malnutrition and diet-related illnesses such as diabetes and heart disease.

Usually, the solution has been for the government to provide public goods, known as direct provision. They use taxpayers' money to fund the provision, spreading the cost over a large number of people who would not be prepared to pay individually. Another innovative approach has been public-private partnerships. The government might work in partnership with the private sector, providing the financing for the establishment of the goods, and then allowing the private producer to run it. This is often more efficient than the government doing it since the private sector would specialize in producing a certain good.

## Market Power

Another example of market failure has to do with corporate market concentration. Over the past 20 years, market concentration has grown significantly in food production, transportation, retailing, and agrochemicals. In 2001, seven companies controlled 90% of global agrochemical sales. For example, Monsanto's acquisition of Seminis, a leading producer of conventional vegetable seeds, has given it significant control over global seed markets, both genetically modified and conventional seeds. Significant market power is considered a market failure because firms with market power may not produce at the lowest possible cost due to a lack of competitive pressure to minimize costs. This results in higher production costs and prices than in a competitive market. Monopolists produce less than the socially optimal quantity of goods, creating a deadweight loss where potential gains from trade are not realized.

Subsidies to agricultural corporations have resulted in substantial negative externalities, leading to environmental degradation, higher health expenses, and the disruption of rural communities. There is consensus that the elimination of agricultural subsidies will improve the global food trade system; however, the negotiations are complicated, considering the huge involvement of powerful market players and lobbying pressures. These highly powerful groups of market players strongly influence the upkeep of advantages in the food systems and further hinder progress toward achieving any sets of equitable international agreements.

Within countries, strategies to combat monopolistic market failure may involve price regulation that has a price ceiling that controls the prices that exist in free markets. They may also use more anti-trust laws that ensure fair competition and eliminate any unfair business activities that may have adverse effects on consumers. Likewise, state ownership can ensure that essential services and goods become accessible to all kinds of citizens regardless of their economic status. This reduces the risk of private companies exploiting consumers through high prices or restricted access.

Internationally, addressing market concentration necessitates involvement from organizations like the UN and WTO, integrated into global accords. Key considerations include safeguarding property rights for the disadvantaged, such as farmers and holders of traditional knowledge, fostering collaboration among poorer nations (e.g., through the G20 in WTO negotiations) and within nations (e.g., via producer cooperatives and labor unions), supporting non-profit scientific research relevant to the needs of developing countries, ensuring transparency and information access, and empowering poorer countries and groups to utilize available information effectively.

In summary, Rocha talked about many sources of market failure. Negative externalities like pollution and public health issues are most prominent. Similarly, corporate market concentration and power has often led to market failure due to abuse of monopoly power. In addition, the lack of regulation in the use of common resources like fisheries and forests has led to the depletion of such resources. Lastly, the inability of unregulated markets to provide public goods such as food security leads to a suboptimum market situation. The challenge is always to identify which strategies will most effectively achieve food security and fulfill the right to food. This often consists of a process of empowering or establishing market institutions and encouraging competition, and/or devolving service delivery.

One of the central concerns in the practical application of Pigouvian taxes within the food industries in order to address market failures—as identified by the paper Lusk (2013)—is the inability to quantify the exact amount of social costs that these industries impose onto the environment. Examples of such food production practices include pollution from agricultural runoff or GHGs from livestock. Lusk comments that, without correct and steady estimates, determination of the "optimal" Pigovian tax becomes significantly impossible.

Lusk draws much attention to the issue of transaction costs and, respectively, the possibility or impossibility of negotiation or even litigation as an alternative to taxation. Coase's theorem suggests that if property rights are clearly defined and transaction costs are low enough, then private parties can, through negotiation, settle issues of externalities without the intervention of the government. In food production, though, where a number of parties are often involved and affected—like different farmers, processors, and consumers, to say nothing of environmental stakeholders—transaction costs will be too high for private negotiations to resolve the externality problem. The political economic effects of the Pigovian taxes are also considered.

He warns that the theoretical elegance of those taxes usually keeps underestimation constant about the tricky ways of its political implementation. Drawing on Acemoglu and Robinson (2013) and Buchanan (1969), Lusk indicates that political capture—or interaction-of Pigouvian taxes with pre-existing market distortions like subsidies or regulations—can yield unintended consequences. This is a particular problem for agriculture, which is already strongly susceptible to the influence of lobbying and interest groups, and where policies can often create policy outcomes that exacerbate identified externalities rather than ameliorate them. Lusk's work has been an important critique of the current discussion in terms of the externalities of food production. He further emphasized that any application of the Pigouvian taxes must be more cautious, evidence-based, considering the complicatedness of measurement of the external costs, possible alternatives, and even risks of political manipulation.

His work, therefore, requires balance in perspective that takes full account of both the limitations and possibilities of using economic instruments to tackle the hidden costs of modern food production. In so doing, Lusk contributes to a more-informed debate on the nuanced role of government in the regulation of externalities and presses for those interventions that will genuinely enhance social welfare and not for ideological or political ends.

Steier's (2011) research examines specific instances of market failure, with a focus on externalities associated with industrial food production. Documentaries such as *Food, Inc.* and *The Future of Food*, along with notable films like *Fast Food Nation*, expose unsettling realities pertaining to the food sector that do not garner public interest as effectively as advertisements from major fast-food corporations like McDonald's or Burger King. These films highlight the fact that consumer choice has profound implications in many areas, from ecological destruction to cruel treatment of animals, often masked by misleading labels and advertisements. It is arguable that consumers are contributing to a kind of "diet of conscientious inconsistency." Many consumers tend to favor taste more than ethical and environmental considerations, often influenced by advertising that promotes harmful eating practices.

This enables an affective disconnection from the negative impacts involved with their food choices, in a way that is analogous to how food companies shift the production costs. Lack of concern for the environmental impact of producing food is a form of willful ignorance. It is well documented that to produce 1 kg of high-grade animal protein, approximately 6 kg of plant-based calories must be supplied to the animal, leading to significant losses in energy conversion. Consequently, there is a need for a critical interrogation of the regulations and subsidies that cause this anomaly by citizens, especially influential individuals such as lawyers and legal academics. The cultural preference for cheap and palatable food is deeply embedded in habits shaped by advertising campaigns rather than biological preference for taste. Industrial ways of farming have replaced traditional practices, and food companies now dominate the marketplace as farmers try to lower their food costs and meet growing demand.

The result is an over-abundance of food produced through industrial agriculture, in which a few large companies control much of the marketplace. This makes it difficult for consumers to choose sustainable options because organic and sustainably produced products are not as widely available or presented compared with their factory-farmed counterparts. In order to change this, the government has to get involved, but change will only be achieved if enough people become aware of the issue. Initiatives similar to those that successfully promoted anti-smoking campaigns can thus be used to promote organic farming and better dietary choices. National campaigns, such as the Food Pyramid campaign, could also work to correct the unbalanced view of nutritionally poor foods etched in customer minds by fast-food and beverage companies. Additionally, such campaigns should promote vegetarian and vegan diets as viable, attractive, and environmentally compatible options. Such efforts could encourage healthier eating habits and reduce the environmental impact of food production. This would involve not promoting unhealthy foods through coupons and toys, while healthier options should be made more attractive and better marketed.

A shift toward sustainable farming and healthier eating habits can help correct market failures and reduce the negative effects of current food production practices through public education and awareness. In the United States, large corporations dominate much of the poultry and livestock products. Their names are probably familiar: Archer Daniels Midland, Tyson, Perdue, Cargill, Smithfield, Land O'Lakes, Kraft, Nestlé, General Mills, PepsiCo, and Dole. These companies foster a commercial environment that strengthens their own positions through exploiting lenient regulations and lax pollution controls to further their dominance in the marketplace and shift the negative consequences, such as ecological degradation and threats to public health, onto others. Factory farming—a primary factor behind the corporate food industry's success—represents a sharp contrast to the idealized picture of old traditional farms. It is an expression of logic that focuses solely on efforts to minimize production costs while disregarding any broader impacts, including environmental degradation, public health hazards, and animal cruelty. According to the author Jonathan Safran Foer, factory farming considers the natural world as an impediment to be managed and anything deemed a cost that can be externalized is suitable. This practice is condemned as full of profit, without any concern for sustainability; it usually leaves a heavy ecological footprint, most explicit in the production of fast food by using hamburgers. The fact that people do not know and do not care about finding out where their food is from is another reason why the spotlight given to factory farms allows it to continue to operate.

### Market Failure Caused by Advertising

One of the major contributors to market failure is consumer manipulation through advertising strategies. As the food industry expanded, companies began to use new advertising techniques that would shift consumers' preferences toward their products over their competitors'. These promotional campaigns often exploited the hardwired human response of salivation upon the sight of appealing foods and became increasingly more aggressive marketing strategies. The food itself became intellectual property. Signs were applied to kinds of food, like nuts, grains, or even meat. The first U.S. registered trademark for food was awarded in 1868 to Underwood Deviled Ham. This protected the brand identity and moved what people ate toward more market-driven decisions.

One of the notable cases in food trademark law was that of "Mr. Peanut," when Planters Nut & Chocolate Company tried to trademark a peanut-shaped character. These and similar cases reveal how food marketing, done through identifiable mascots or jingles, influences consumer preference at times more strongly than natural choice. This sort of marketing often relies upon "classical conditioning," whereby the invariable linking of a brand or product with a positive experience—such as a catchy jingle or a much-loved mascot—results in an unconscious leaning for the consumer, especially children. The pervasive nature of food marketing has confused what a food product is and what entertainment is, often leaving consumers unable to discern how they are being influenced. Calls for awareness about these practices, especially when it involves children, usually fall flat since these ads remain in a place to influence eating habits. Take the advertisements for Honey Nut Cheerios, which employ engaging elements to distract from the fact that it is an advertisement at all.

Marketing messages often promote processed forms of food, like peanut butter, over healthier, whole versions. This is demonstrated by how Mr. Peanut linked peanuts to pleasurable social functions rather than emphasizing their nutritious benefits.

Similarly, peanut butter brands like Skippy and Jiff are sold at artificially lowered prices, as if the real production costs do not need to be considered. In either case, the consumer public is enchanted either by cartoon characters or frightened about the safety of their food but rarely considers the true nutritional value of what they eat. This case epitomizes the higher-order failures in the food market, where hidden costs are borne by society due to false advertising and other market aberrations.

### Food Safety and Contamination

Steier discusses the food safety issue and how it is often compromised in industrially constructed food production systems. It highlights that food safety scandals often happen because of problems well entrenched in the systems of industrial food production, such as contamination across huge processing facilities. The problem is compounded further by the fact that standardized seeds are employed in crops, which reduces crop diversity and heightens susceptibility to diseases. The consolidation of food processing activities and the heavy distribution of food products increase the vulnerability of widespread contamination, as evidenced by the peanut contamination incident of 2009.

Steier criticizes the fact that the food industry shifts the costs of foodborne illnesses to society in general, thus, in effect, receiving a subsidy in the form of public health. He reiterates that such a practice creates an economic incentive for the industry to continue dangerous actions, considering that the economic costs of foodborne illnesses are being carried by society, not the producers. This, however, is further demonstrated through the cases of Salmonella and Listeria and the vulnerability of the people to these diseases.

Yet, the system of regulation concerning food safety is seriously deficient, often reactive and rarely proactive. It means that close control, a higher quality of food inspection, and public education about food sources could provide changes in consumer behavior and force an industry towards safer practices. In addition, the discussion would even present the idea that increased awareness by consumers of foodborne illnesses and dangers may empower them to take legal avenues for higher standards in the production of food. The current food industry resists complete development of risk controls in this manner, at least for those pathogens that take longer to incubate, due to perceived costs. It is indicative that strict controls and internalization by the industry of those costs would be a better way to deal with the negative externalities associated with food safety.

Elmes (2016) reviews how economic disparity affects food security in the United States, giving special attention to the ways inefficiencies in the food market enhance the problem. In the United States, economic inequality has reached levels not previously witnessed; between the mid-1980s and 2012, the percentage of wealth controlled by the bottom 90% of the population shrank from 35% to 23%, while the percentage of the wealth owned by the top 0.1% grew from 7% to 22%. This growing income inequality has caused significant increases in food insecurity, particularly among lower-income groups whose accessibility to affordable, healthy food options is severely limited. Due to the strained economic conditions, many low-income earners are forced to consume cheaper and less nutritional food because of which health deteriorates; this therefore reduces their capability to be integrally a part of community activities.

In a USDA report released in September 2015, it was found that 14 percent of American households, were food insecure in 2014. The same report also noted that in 2014, compared to metropolitan areas, food insecurity was higher in rural areas and food insecurity rates were 26.1 percent for Black households and 22.4 percent for Hispanic households. Studies have also demonstrated that food insecure individuals are at increased risk of experiencing hunger and consuming more high-calorie, nutrient-poor foods in an attempt to feel full. As economic resources become scarce, especially towards the end of the month, people may either skip meals or also resort to nutritionally inadequate diets lacking certain basic products like milk, vegetables, and fruits.

Foods that represent the dietary patterns consumed most frequently are filled with fast and processed foods high in refined sugars, saturated fats, and salt-economic accessible that are nutritionally poor and highly marketed within the industrial food system. This system alone, based on maximizing profit through the distribution of processed foods, is a great contributor to worsening food insecurity and obesity. Also, the market share held within the food industry increases these problems further. In the last twenty years, both vertical and horizontal integration have concentrated the marketplace into a few corporate giants commanding high market shares, reduced competition, and higher costs for nutritious food. This vertical integration enables corporations to dominate various levels of the supply chain, sometimes right from production to retail, for better efficiency in costs and profitability.

Meanwhile, horizontal integration involves buying competitors to increase market share. This consolidation has reduced the economic influence of the smaller producer and consumer, leading to an oligopoly where a few firms, such as Walmart, use their market dominance in price and terms dictation. From 1990 to 2011, the market share of dominant food industries increased significantly in concentration: beef slaughter rose from 69 to 82%, pork slaughter from 45% to 63%, wet corn milling from 74% to 87%, and soybean processing from 61% to 85%. Similar patterns of concentration are witnessed in the international fertilizer industry and the grain trade, where a few companies have an enormous share of the market. This concentration of power within the



aforementioned institutions has subsequently granted them leeway to focus on short-term profits rather than social benefits by creating unequal access to healthy food. Elmes blames the industrial structure for this food market failure because it prioritizes profit over public health, environmental viability, and equitable distribution of food.

Such concentration of economic power, supported by convenient agricultural policies and subsidies, has made the U.S. food system resistant to transformation and perpetuates existing inequities. It has motivated a host of social movements and creative initiatives in which Slow Food, Food Sovereignty, and community food security work toward democratizing access to healthy food while reducing harmful effects associated with industrial agriculture and capitalist logics. Elmes proceeds to make clear that nutritional food is an important aspect of the total effect in participation in life and economy. In his opinion, it is impossible to have health, education, productive work, and social participation without food.

The current industrial food system, typified by market concentration and profit-maximization goals, reduces these abilities and perpetuates poverty, food insecurity, and obesity among the economically disadvantaged. This systemic issue has brought about demands for regulatory measures and grassroots movements that promote the right of every individual to have access to healthy nutrition.

Collart and Interis (2018) study the phenomenon of consumer uncertainty concerning date labels on food products and implications for food waste. According to their findings, the consumer often does not have enough information at the point of purchase, leading to the disposal of items that are still consumable due to misinterpretations of date labeling.

This research investigates the possible effect of the provision of additional information related to date labels and ecological consequences of food waste on consumer decisions involving foods with different shelf lives and those approaching or which had exceeded their "best-before" dates. The findings revealed that mere consumer education on the meaning of date labels did little in changing their willingness to purchase foods that were past their best before dates. Participants in both the control group and the group that received basic label information consistently preferred foods labeled as unexpired over those labeled as expired, regardless of the product's perishability. However, the receipt of extra information regarding environmental consequences of food waste—LABELS+ENV treatment—started to change their preferences. The most striking changes concerned expired frozen products—Expired 1 day and Expired 9 days—and recently expired semi-perishables—1 day expired. The subjects in this group were generally indifferent between the expired and unexpired frozen foods, while they were mainly indifferent—78% between unexpired semi-perishables and those that had expired just a day before.

The study also observed significant variation in individual preferences for expired food products. The individual-level willingness-to-pay data supported overall findings for expired frozen foods and recently expired semi-perishables, yet provided further nuance in terms of how consumer preferences shift across treatments for expired perishable and semi-perishable products. Consumer attitudes did not just shift for expired products. The study would also point out changes in preferences for unlabeled, non-expired frozen foods within the LABELS+ENV treatment. Their results point toward an information gap between date label confusion, food waste, and the resulting environmental impact. Filling this gap would, in turn, help policymakers and/or food retailers raise consumer willingness to purchase expired products depending on product perishability.

At home, if consumers are indifferent between choosing unexpired and expired food items, they may opt to consume expired items rather than discarding them to purchase new ones. Similarly, in stores, if consumers show no preference between expired and unexpired items, then retailers would be less motivated to remove expired products from the shelves, hence reducing waste. This study also identified that the most significant changes in the consumer behavioral patterns due to the environmental information was about frozen foods. The reason could be that the consumers feel that eating expired frozen foods is less risky regarding food safety or quality than the use of expired perishable or semi-perishable items.

Other studies also note that some people freezing products of food also regard freezing as a good preservation method and disregard the shelf-life date. That the effect of the LABELS-only intervention was so small suggests that, if industry stakeholders and policymakers are to effectively reduce expired product-related food waste, clearer date labels will not be enough, but rather stronger incentives for changing consumer behavior. This may involve public awareness campaigns or marketing strategies that highlight the environmental and economic benefits of reducing food waste. Current strategies to reduce the problem include policy action based on standardizing and educating date labeling. Examples of such strategies in the United States include the passing of the Food Date Labeling Act of 2016, as well as voluntary standards on date labeling developed by both the Food Marketing Institute and the Grocery Manufacturers Association. At the European level, the European Platform on Food Losses and Waste has within it a subgroup dedicated explicitly to date labeling issues.

While initiatives such as ReFED have identified consumer education and standardization of date labels as two cost-effective strategies to reduce food waste, the results of this study suggest that even when consumers understand what date labels mean, their preferences for expired versus unexpired foods are unlikely to change greatly. Therefore, a broader approach that incorporates environmental awareness with economic incentives may be necessary for effectively reducing food waste and its associated impacts.

A report issued by the OECD Library suggested that farmland subsidies and price supports in Mexico have created market distortions. These governmental programs to help poor, small farmers often result in overproduction of certain crops, while sometimes neglecting other, more viable or demanded crops. This misallocation of resources is a type of market failure since the government policies inadvertently promote inefficient means of production as opposed to allowing market forces to allocate resources in the most efficient manner possible.

This report also deals with another very critical type of market failure-asymmetric information-which is especially relevant to the rural and dispersed areas in Mexico. Many thousands of small-scale farmers have no or very little information on prevailing market prices and the best agricultural practices, and even less on technology, for making an informed decision. Hence, non-access to information makes them less competitive and ineffective in farming methodologies, leading to vicious circles of poverty and low productivity for these small-scale farmers.

The report identifies various formidable constraints to market access of small-scale producers in Mexico, a fundamental factor that tends to accentuate market failure. These constraints include poor infrastructure, represented by bad road conditions and a general lack of transportation options, raising transaction costs and reducing market efficiency. Smallholders often have difficulties in accessing credit and financial services, both an important investment to be made in technologies, boosting productivity.

Furthermore, much market power is held by a few large-scale agribusinesses, creating a number of inequities and inefficiencies in the food market. Major players have the potential to control market circumstances, manipulate the price mechanisms, and even influence policy decisions to self-serving ends. This unequal power distribution thus translates into the market levels where small-scale farmers have to accept lower prices for their produce or products while consumers pay a higher price; this creates dual market structure and therefore, inefficient resource allocation.

Marcone et al. (2020) highlight that market failure in Canada's food industry is characterized by widespread food insecurity and disparities in access to nutritious foods, particularly affecting low-income households, indigenous communities, and rural populations. According to the FAO, food insecurity refers to "a situation when people are without safe, sufficient, and nutritious food and are unable to attain it on a regular basis". Food insecurity in Canada afflicts 1 in 8 households, over 4 million Canadians, encompassing 1.15 million children. Low household income, unemployment, homelessness, and very high costs of living - encompassing chalking up huge bills for food and medication-are particularly pervasive influencers in this dire situation. The consequences of food insecurity are serious, manifesting in poorer physical and mental health outcomes: obesity, malnutrition, anxiety, depression, and other chronic diseases. Already vulnerable populations include recent immigrants, the underemployed, elderly persons, and indigenous groups whose economic or geographic circumstances limit accessibility to healthy food options.

The high contribution of economic barriers to the worsening food insecurity in Canada is very much real. Though Canada's food markets are relatively more affordable than most developed economies, wide disparities in affordability and access to adequate amounts of nutritious food persist across socioeconomic groups. Many low-income households often experience a high cost of foods, a limited variety, and are especially reliant on foods that are cheaper but more energy-dense and nutritionally inferior. This problem is further exacerbated by the proliferation of "food deserts"-urban areas with no accessible, reasonable fresh food available anywhere at reasonable prices. The necessity of using low-cost, high-calorie foods, like fast foods and processed snacks, further promotes what is described as an "obesity paradox," where the least-advantaged populations have a greater risk of obesity despite their food insecurity. These products are more affordable and accessible than nutritious foods, which are mostly costlier and less available within low-income neighborhoods and rural settings. Geographical disparities further complicate food access in Canada, particularly in rural and remote communities.

Residents in these areas, especially in the northern regions, face additional challenges related to food availability and transportation costs. Fresh and nutritious foods are often scarce or unaffordable due to the high cost of transporting these items to remote locations. Consequently, local stores and food outlets tend to stock non-perishable items over fresh produce, meats, and dairy, further limiting healthy food options. That means Indigenous peoples-who already face restrained income opportunities and higher food costs-are disproportionately affected. While various initiatives, such as community freezers and sharing food in a collective manner, have been implemented to help overcome these challenges, deep-seated systemic barriers persist, making food security a continued struggle. Several public policies and programs exist in Canada to address food insecurity and to promote better nutrition.

These include a revision of Canada's Food Guide, nutrition labelling of packaged foods, and moves to regulate food marketing to children and other vulnerable groups.

Yet, there are some limitations to these policies. For example, while it was widely emphasized to reduce sodium and trans-fat levels in processed foods, the inadequate regulation with respect to advertisement of food would let people continue with advertisement of high calorie, low nutritious foods. Initiatives on food price modification and their affordability too didn't address the economic imbalance between healthier and less healthy options good enough to keep the low-income families away from healthy diet options. Social and cultural factors affect dietary practices as well as bring inefficiencies along the food value chain. Family dynamics, traditional culture, and mass media have a significant influence on consumer choices. Children and adolescents, in particular, are highly vulnerable to marketing and promotion activities that suggest unhealthy food options, leading to unhealthy eating and increased rates of childhood obesity.

This is especially true for facilitating healthy eating at school and through the system of social support, such as programs for seniors and low-income families. However, larger economic and environmental factors often restrain the potential influence of these interventions to assure adequate availability and choices of food. Marcone et al. propose that biotechnology presents a promising pathway for tackling various challenges linked to food insecurity and market failures within the food sector. Developments in genetic engineering have facilitated the cultivation of crops that exhibit greater resilience, including resistance to drought and diseases, as well as enhanced nutritional profiles. Such innovations possess the capacity to augment the efficiency of food production and lower costs, which may ultimately enhance food security. However, at present, the impact of biotechnology is limited by regulatory frameworks coupled with public acceptance challenges that make comprehensive approaches to technological advancement critical in addressing food insecurity.

Deconinck (2021) considers that the interaction of market concentration and buyer power are critical in determining market failure. The structure of the food supply chain is basically "hourglass-shaped": many farmers supply their products to relatively few processors and retailers, who sell them to a large number of consumers. In this kind of structure, huge inequalities in bargaining powers can arise; in other words, processors and retailers may have strong market power against farmers and other suppliers. As a point of fact, this level of concentration is measured using the Hirschman-Herfindahl Index, or HHI, which suggests that "many segments of food processing exhibit significant to high levels of concentration, such as specialty canning and snack foods." Because of this, farmers often are forced into a structurally disadvantaged position, selling their products for lower prices than would be expected under more free market circumstances.

Deconinck also mentions that Unfair Trading Practices (UTP) are common within the food supply chain, exacerbating the undue market imbalance in power. These practices include late payment to suppliers, forced rebates, unilateral changes to contract terms, and any other shifting of costs onto farmers and small-scale operators. For instance, buyers would ask for late payments or a revision of the contract terms after the transaction has already been agreed upon, which would shift more costs onto suppliers or force them to accept less favorable terms. Such UTPs generate uncertainty and financial stress for farmers, who might be in a weak bargaining position, thereby reducing their capacity for investment in productivity enhancements or innovation. Another pertinent issue raised relates to the incomplete transfer of prices, effective from the agricultural sector, into the retail sector.

The study explains that the change in agricultural prices does not always correspond to changes in consumer prices, which suggests price in transmission. This situation is often used as an indicator of un-competitive market behavior when large firms in either processing or retail marketing are able to manipulate prices for their own advantage. For example, even in situations where agricultural prices fall, retail prices may not be correspondingly lowered, allowing retailers to boost their margins. In return, it would mean that when farm prices go up, the retail prices could increase disproportionately. Price-setting behaviors in this way would suggest dominant firms are exerting market power at the expense of consumers who pay higher prices and farmers who receive lower product prices. Due to their substantial purchasing power, big businesses can often place pressure on suppliers, which in turn can drive up the prices of agricultural products to levels below those that might have prevailed in a more open market. This market power can dampen investment and the diffusion of innovation at the farm level, as lower profits make it tough for farmers to finance improvements or adopt new technologies.

Such dynamics may therefore cause market failures in which the competitive equilibrium is not realized and, therefore, this benefit of competitive pricing cannot be felt down the supply chain. In addressing such market failure, Deconinck calls for more active policy that includes monitoring and enforcement such as the introduction of obligatory codes of best practice to ensure fair trading practices. It improves the transparency in the mechanism of price determination and supply chains to reduce negative consequences of market concentration and buyer power. The potential effect of such policy measures is that the balance of power in the food



supply chain is redistributed, hence ensuring more equitable outcomes for farmers and consumers, with the development of a more competitive and efficient market environment.

### Meta-Analysis

This meta-analysis will contrast the following sub-topics: Geographical, Industry-Specific, Market Structure, and Policy.

### Geographical Comparisons

The food industry in the United States has characteristic industrialization features that come with a lot of environmental externalities. Agricultural practices apply a huge number of synthetic fertilizers and pesticides, especially in regions like the Corn Belt, leading to critical problems such as soil degradation, water pollution, and emissions of greenhouse gases. The United States also experiences severe information asymmetry in the processed food market, where marketing practices often hide the true nutritional value of the merchandise. Market power is also heavily concentrated, with market control in the hands of a few firms. This leads to a properly oligopolistic structure in which effective competition is stifled, especially regarding meatpacking.

Canada, while also facing similar challenges environmentally with farming and animal husbandry, operates in different regulatory conditions-particularly with its supply management system concerning dairy and poultry. This stabilizes farmer incomes, but at a very real cost: higher consumer prices and reduced market competition. Despite Canada implementing more stringent labeling regulations than those in the United States, consumers continue to face information asymmetry, especially in differentiating between marketing terminology such as "organic" and "natural." Typically, consumers do not recognize the distinctions between these two concepts. Furthermore, the consolidation of market power within the supply-managed sectors restricts the entry of new competitors. Collectively, these factors contribute to a distinct set of market dynamics when contrasted with those in the United States.

Agriculture in Mexico presents a mix of more traditional small-scale farming and large-scale industrial farming, each with its set of associated externalities. For instance, the state of Sinaloa is recognized for large-scale farming of vegetables and fruits, but it also confronts problems like the overexploitation of water resources and pesticide runoff, which increase water scarcity. Information asymmetry is more marked in rural areas, where lower literacy rates and less access to information impede the ability of consumers to make informed choices. While the market power in Mexico is less concentrated than in the U.S. and Canada, there are still disparities, with large agribusinesses being much larger than small farmers; this results in a lower price to producers and higher prices for consumers.

### Industry-Specific Comparisons

Agriculture across North America is a significant source of environmental externalities, although the extent and type of impact vary. Both within the United States and Canada, the focus on monocultural practices, along with the use of synthetic inputs, leads to soil degradation and loss of biodiversity. Water use in agriculture, as on almonds and alfalfa in California, has also led to critical water shortages, further elevating concerns about sustainability. The problem of information asymmetry within the agricultural sector, especially with respect to the application of GMOs, increases the complexity of the issues concerning market failures, as misconceptions at the public level influence consumer behavior and regulatory mechanisms. This is also exacerbated by the dominance of few firms in the seed and pesticide markets, which ultimately raises farmers' costs and reduces crop variety.

The processed food industry has a huge impact on the creation of public health externalities throughout North America: the rising availability of high-calorie, low-nutrient products has caused an increase in obesity rates and health complications related to it. Of particular concern is the United States, where fast food and sugar-sweetened beverages are critical elements in the crisis of obesity. Information asymmetry remains high in this sector, despite efforts to increase transparency through measures like the FDA's Nutrition Facts label. Too often, consumers are unable to understand the nutritional facts presented and, therefore, do not make the best choices regarding their nutrition. Second, the processed food industry is highly concentrated, where a few multinational companies dominate the market, exercising much influence over consumer tastes and the operations of their suppliers, often at the expense of product quality and sustainability.

On the other hand, there is food waste in North America at the levels of distribution and retail, which is related to aesthetic criteria and consumer purchasing habits. This waste represents a significant economic loss and contributes to environmental problems, such as the release of methane from landfills. Retailers further exacerbate information asymmetry by promoting products with unclear claims of being "healthy" or "sustainable," without providing clear standards to back them up. The concentration of market power in just a few large supermarket chains allows them to exert strong pressure on suppliers, leading to strategies that may compromise the quality of products and limit choices for consumers.

## Market Structure Comparisons

The structure of food markets in North America—whether described as competitive, monopolistic, or oligopolistic—has enormous implications for the nature and pervasiveness of market failures. In competitive markets, particularly within agriculture, the pressure to drive down costs often results in negative spillovers to the environment, as the small farmer overuses fertilizers or pesticides in attempts to maximize short-run yields. This may result in the long-term degradation of the soil and the pollution of waterways. Competitive markets are similarly characterized by information asymmetry in that firms will employ assertive marketing strategies to further influence consumer behavior in food, especially processed.

Monopolistic markets are the ones where leading firms disregard sustainability for profit maximization and, as a result, create significant negative externalities. For instance, a monopolistic food processor may source its supplies from firms that contribute to deforestation and face no consumer dissatisfaction as alternatives are not available. Moreover, monopolistic organizations have resources to shape public opinion through widespread marketing and lobbying efforts and enhance information asymmetry. Because there is an absence of competition, firms in such markets may determine price while disregarding consumer welfare or even the equity of the supplier, and by doing so, it results in high prices, lessened innovation, and exploitation of suppliers, particularly in areas where regulations are weak. Oligopolistic markets, where just a few large firms dominate, can result in collusion or tacit coordination that worsens market failures.

The few major players wield large market shares in North America's meatpacking industries and hence have greater market power, enabling them to set up practices that harm the environment, such as overproduction and waste. Information asymmetry also characterizes these markets, in which dominant companies alone have the wherewithal to manipulate consumer perceptions through advertisements and branding. The prevalence of market power constrains competition and diminishes consumer options, often leading to higher prices and decreased product quality.

## Policy Comparisons

Rocha advocates interventionist policies such as imposing pollution taxes, setting regulatory caps on the number of animals per factory farm, and enforcing strict waste management practices. These measures are aimed at internalizing the external costs of pollution and environmental degradation caused by factory farming and intensive agriculture.

Lusk supports the idea of Pigovian taxes to address negative externalities but emphasizes the practical challenges in implementing them, such as accurately estimating social costs and navigating political complexities. He rather proposes that evidence-based methods be followed to ensure that taxes internalize the externalities with minimum chances of unintended outcomes. Steier's policy recommendations center on enhancing public consciousness and education regarding the environmental consequences associated with industrial food production. He advocates for national initiatives akin to anti-smoking campaigns that aim to encourage sustainable agricultural practices and healthier dietary selections. In contrast to Rocha and Lusk, Steier does not prioritize direct economic interventions; instead, he concentrates on modifying consumer behavior via heightened awareness.

## Discussion and Conclusion

The meta-analysis of the literature on market failures in the food industry reveals that all four primary sources of market failure—externalities, information failure, market control, and public goods—are interconnected. Though most of the individual studies have focused on specific issues, environmental externalities, or information asymmetries, the overall analysis explains how these market failures can be connected with each other and further aggravate each other. For instance, the dominance of big businesses with considerable market power biases not only market governance but also deepens information asymmetry through labeling and marketing discourses. This interrelation would mean that even a single-point intervention, such as increased regulation or improved labeling policies, would not be sufficient. What is called for is an integrated approach that links regulatory structures and public funding of infrastructure and R&D with market-oriented incentives, such as carbon pricing, if serious action on the part of environmental challenges is to be taken seriously.

The meta-analysis also quantifies the wide social repercussions of market failures, mainly their contributions to enhanced social inequities and food insecurity. Health-impacting externalities, like those from pollution or the consumption of processed foods, are taken to a very great degree by low-income communities with limited access to healthy nutrition options. Similarly, the market power of large firms can limit both the availability and affordability of diverse food choices, especially within disadvantaged areas, which further exacerbates disparities. Therefore, food insecurity requires a bifocal response that strikes a balance between economic efficiency and social equity; hence, the need for policies that advance inclusive economic growth and equity in resource distribution. The literature also highlights multisector involvement in food insecurity.

Policies with horizontal integration of programs in various sectors like health, education, agriculture, and social services have more synergistic and reinforcing interactions. This may result in more harmonious and durable food security policies that can address the economic and social dimensions of the problem. Once more, the need for political will to that cause and awareness of the right to food from the citizens themselves should be underlined. Advocacy and building public awareness are important in creating the required political will for developing holistic food security approaches. Food insecurity has interlinked economic and social dimensions; making the public aware of this fact may create demand for better interventions and hence a favorable political climate for far-reaching reforms. Besides, the meta-analysis identifies possible ways innovation and technology can solve inefficiencies in the market. Improvements in blockchain technology, for example, could enhance transparency in the value chain and thus lower information asymmetry.

Similarly, plant-based proteins and agricultural production techniques could enhance reductions in the ecological footprint and public health externalities. Nevertheless, these technological changes and political measures would strongly depend on social acceptance and consumers' level of education. Such consumer practices, not to mention cultural reasons, have been among the major causes of perpetuation of market failures; this, therefore, implies that the economic and regulatory interventions need reinvigoration with cultural change and greater involvement by the public.

### Recommendations for Future Research

Future research should focus on such interdependent features of differing forms of market failure in the food economy, such as how market power, information asymmetry, and environmental spillovers feed into each other. Understanding the interlinkages will help shape more coherent and effective policy initiatives aimed at addressing several failures at once.

The development of dynamic models, which are able to simulate the long-term impacts of various policy interventions regarding carbon taxation and the promotion of sustainable behaviors throughout the whole food supply chain, from production to consumption, would be very useful. Such models have to be able to include adaptive behaviors of the different agents Farmer, retailer, consumer-and potential feedback that might emerge in order to help policy makers perceive the broad systemic outcomes of their actions.

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## 1.5: Post-Secondary Academic Performance of Hispanic Students- A Review of the Literature

**Title:** Post-Secondary Academic Performance of Hispanic Students: A Review of the Literature

**Author:** Katie Zheng

Standardized examinations were employed by Duran (1983) to assess Hispanic students' academic performance in college classroom environments. The report's findings highlight the importance of examining and comprehending the variables that affect Hispanic students' performance in postsecondary education. This includes tackling issues related to educational attainment, college readiness, and sociocultural effects. The article discusses problems and prior research on the prediction of achievement for Hispanic students in the United States. It mostly summarizes empirical studies and concludes with recommendations for additional research to be conducted in order to enhance college performance.

After providing a general demographic overview of Hispanics in the United States, Duran analyzes national surveys of the Hispanic population as a whole and of particular Hispanic subpopulations. He then focuses on population size, ethnic subgroup, age, economic welfare, language background, and prior educational attainment. This gives the reader a broad overview of the educational and demographic characteristics of Hispanics in the US population at large, with a focus on Mexican Americans, Puerto Ricans, and Cuban Americans as the main subgroups under consideration.

Duran follows the demographic and educational attainment overview of Hispanics in the U.S. by briefly summarizing recent major studies on Hispanic students' learning achievement during high school. The goal is to provide a concise summary that offers general insights into Hispanic students' academic performance before they reach college entrance opportunities. This summary highlights many obstacles Hispanics encounter in becoming academically prepared for college. Additionally, it discusses the relationship between Hispanic students' economic, social, and linguistic backgrounds and their high school achievement, drawing on findings from national surveys and other research.

Following the discussion on high school achievement, Duran examines the characteristics of Hispanics' college admissions test scores using recent data from the American College Testing Program, the College Entrance Examination Board, and the Graduate Record Examination Board. The primary aim is to detail the test score performance of Hispanics, including breakdowns by specific subgroups where possible, and to compare their performance with that of non-ethnic minority examinees. Additionally, a secondary but limited goal is to explore the relationship between admissions test scores and the language characteristics of Hispanic students, based on their responses to questionnaires completed when applying to take these tests.

The study reviews empirical studies that evaluate the ability to predict undergraduate Hispanics' college grades based on high school grades and college admissions test scores. Due to the scarcity of publicly available and published studies, the research primarily focuses on Mexican American students. Additionally, it discusses a significant series of studies conducted on Puerto Rican students by the College Board Puerto Rico office, which used high school grades and college entrance exam scores to predict college GPA. These studies, along with one conducted in Mexico, are noteworthy as they suggest that high school grades and admissions test scores effectively predict college performance in contexts where Hispanic culture and the Spanish language are predominant.

Additionally reviewing selective findings on Hispanics from various studies of college aptitude and achievement test performance, the cited findings focus on issues of test bias and the impact of language background and personal characteristics on test performance. This section also includes a small number of studies that specifically examined the role of English proficiency as a moderating variable in assessing Hispanics' college aptitude using Spanish or English test scores.

Duran concludes with issues related to improving the prediction of Hispanics' college grades and other measures of college success. The chapter emphasizes the need to broaden the types of indicators used to predict Hispanic students' college aptitude or achievement. The final portion of Chapter 6 outlines directions for needed research, focusing on three main areas: improving understanding of Hispanics' preparation for college, examining how the college admissions process operates for Hispanic candidates, and exploring the college experiences that influence Hispanics' academic success and the prediction of this achievement.

Crisp and Nora (2012) address the phenomenon that although Hispanics are quickly becoming America's largest and youngest minority group, a large percentage of the Hispanic population faces challenges in higher education. Although the enrollment of

Hispanic students in institutions that grant degrees was expected to rise 38 percent between 2007 and 2018, the percentage of these students obtaining bachelor's degrees started to lag behind when compared to other demographic groups.

From the past 25 years, much research has documented the disparities Hispanic students experience, which includes a lower graduation rate in high school, a lower college enrollment rate, and difficulty in completing degrees. To accompany this, there have been theoretical and conceptual frameworks to accompany and challenge societal perspectives pertaining to unfair portrayals of Hispanic students as a group that lacks ambition or devalues the importance of obtaining a higher education. These studies have been valuable in disrupting stereotypes and racial profiling, but there still is a need for methodologically and theoretically rigorous studies, both qualitative and quantitative for these students.

Crisp and Nora explain that researchers must shift towards producing empirical evidence that can inform effective strategies and programs for Hispanic student success. Existing research indicates correlations between various college experiences and student outcomes among Hispanics, but establishing causal relationships remains a challenge. Given the difficulty of random assignment in educational settings, alternative methodologies such as propensity score matching are crucial for rigorously assessing program impacts. Furthermore, there is a call for developing and testing theoretically sound statistical models that can elucidate indirect and mediating pathways between college experiences and Hispanic student outcomes.

Longitudinal databases that integrate state and institutional records from kindergarten through graduate school are essential for tracking Hispanic students' educational trajectories comprehensively. While national datasets like NELS, BPS, and ELS offer valuable insights into cognitive outcomes, there is a critical need to incorporate non-cognitive measures—such as psychological, social, and cultural factors—into these databases. This holistic approach can capture the full spectrum of influences on Hispanic student success, including interactions with peers and faculty both on and off campus.

Current definitions of "student success" predominantly focus on cognitive outcomes, which may not fully capture the diverse experiences and aspirations of Hispanic students. There is a need to expand these frameworks to encompass psychological and behavioral outcomes. For instance, measures of student satisfaction can reflect not only overall gratification but also the culmination of academic and social experiences that shape students' interactions with peers, faculty, and their environment. The report concludes by stating that research surrounding Hispanic students can benefit from models that better reflect the students' cultural backgrounds and differing experiences, which consider the influence of outside support systems like family, faculty, and peers. Developing a culturally sensitive model is important in accurately depicting the experiences and support needed for Hispanic students in order for them to achieve success in an educational setting, abandoning previous theoretical models that often rely on databases that may not fully account for a diversity of student experiences.

Saenz and Ponjuan's (2011) paper regarding the educational outcomes of Latino males presents a pressing issue in the United States, necessitating proactive and comprehensive efforts. Despite various challenges, opportunities exist to improve these outcomes, particularly given the national, state, and regional emphasis on increasing the number of individuals with college degrees and credentials. Similar to initiatives aimed at improving the educational outcomes of African-American/Black males, emerging national movements focusing on Latino males suggest the potential of promising regional and local programs. This literature review explores several funded programs targeting Latino males, providing a blueprint for communities and educational institutions to develop or enhance programs to support Latino males.

The programs discussed in this review share fundamental practices aimed at assisting communities and educational leaders in improving the success of Latino male students. Based on the authors' experience with successful Latino male initiatives nationwide, a detailed blueprint has been developed. This blueprint outlines critical elements to help education and community leaders devise strategies to increase Latino male success, emphasizing the importance of intentional planning and infrastructure development.

The blueprint addresses three main stages: planning and development, resource development and sustainability, and outreach and communication. In the planning and development stage, initial actions include creating awareness through education and workforce data to communicate current trends for Latino males. Developing a network of support involving education, political, and community leaders, as well as business champions, is crucial. Establishing a vision and mission statement or a theory of change for Latino male success, along with identifying short- and long-term measurable goals and objectives, are essential steps in this stage.

Resource development and sustainability involve developing key benchmarks to track progress and designing research-based programmatic activities to meet designated objectives. Securing necessary resources, including human capital and financial resources, is vital for programmatic planning and execution. Ensuring financial and nonfinancial sustainability is critical for the long-term viability of developed initiatives.



Outreach and communication focus on informing stakeholders about policy and practice recommendations. Publicizing evaluation findings to policymakers, practitioners, potential funders, and other influencers is important for securing champions. A diverse and strategic dissemination plan, utilizing various media outlets, is essential for widespread impact.

Several innovative programs focused on Latino male education illustrate the blueprint's principles. At the pre-college level, programs like Fathers Active in Communities and Education emphasize emotional and informational support for Latino fathers, encouraging their engagement with their children's education. Programs such as Encuentros Leadership, X—Y Zone, and Puente Project provide emotional and instrumental support, empowering Latino male students to pursue higher education. The Latino College Dollars Foundation offers instrumental and informational support, providing Latino families with valuable scholarship information.

At the college level, initiatives like Lambda Upsilon Lambda Latino at Cornell University offer emotional and structural support to improve Latino male involvement and persistence. Project MALES (Mentoring to Achieve Latino Educational Success), a joint effort between the University of Texas at Austin and South Texas College, provides emotional support through mentoring relationships with older Latino male peers and role models. The Doorway to Success: Latino Male Retention Initiative at Monroe Community College and The Clave Latino Male Empowerment Program at Union County College offer comprehensive support to increase academic persistence and degree completion.

Given the significant role of family in the lives of young Latino males, schools and postsecondary institutions must develop policies and practices that actively involve Latino families and community members in the educational experience. Latino families should be considered crucial in creating a college-going culture. K–12 schools and institutions of higher education can enhance outreach and services to families and community members. Leaders need to be sensitive to economic drivers that may push young Latinos into the workforce instead of college. Engaging families and communities in fostering a culture of college success early on is essential for sustaining the necessary support for graduating more Latino males.

The alignment between K–12 and higher education is crucial, and local school districts should develop policies providing equitable access to pre-college curriculum aligned with college and career expectations. School boards play a significant role in enacting policies that influence curriculum, resources, and community outreach, further developing policies unique to the Latino community.

Progress in developing rigorous academic standards and assessments must be complemented by academic and social developmental support. These supports include accessible advising and counseling, mentoring, workshops on study skills and financial literacy, learning communities, tutoring, and college and career planning. Providing linked academic and social supports is crucial for helping Latino males reach high expectations and succeed in postsecondary education. Institutions must also prioritize faculty diversity efforts and enact policies promoting the recruitment, hiring, and retention of U.S.-born Latino and other men of color faculty.

Policymakers and practitioners must recognize the financial hurdles Latino students and families face in higher education. Addressing language barriers, cultural capital disparities, and poor information-sharing is essential for improving financial literacy among Latino families. Institutions should provide accurate data on costs, including net price, and be transparent about financial aid packages, student loan debt, and other financial aspects. Publishing clear data on average time to degree, persistence and graduation rates, and job placement rates is important for informed decision-making by Latino students and their families. Financial literacy programs and policies supporting intentional information-sharing and skill development are crucial for equipping Latino/as to make sound financial decisions. Innovative policies, such as loan forgiveness for education graduates, can encourage Latino males to pursue and succeed in higher education. A comprehensive agenda involving researchers, policymakers, public officials, private sector leaders, and Latino families and communities is shown to be necessary for Latino males to succeed along varied academic pathways.

Hispanic Serving Institutions, or HSIs, initially emerged in response to the concentration of Latino students within higher education institutions in the US. First acknowledged in the 1980s, this discovery led to the formal label of HSIs, which are defined by their enrollment demographics. These HSIs are public, nonprofit degree-granting institutions, with at least 25% undergraduate enrollment being Hispanic.

The initial creation of HSIs started in the 1980s, when policymakers and educators alike in both Texas and New Mexico began to recognize the increasing Latino enrollment in educational institutions. Following this, congressional hearings highlighted the concentration of Latino students in a small number of underfunded institutions, which prompted Congressman Paul Simon to introduce legislation to support new Hispanic institutions. This failed, but began the discussion on equity in educational spaces.



Subsequently, the creation of the Hispanic Association of Colleges and Universities (HACU) in 1986 further galvanized support for the concept of HSIs. These colleges and universities were aimed to support institutions with significant Latino enrollment, helping to create the term “Hispanic Serving Institution”. Simultaneously, the *Lulac v. Richards* case in Texas emphasized the lack of funding for institutions serving many Latino students, eventually leading to the South Texas Border Initiative in 1989, providing financial support for nine Texas universities with high Latino enrollment.

On the federal level, HSIs gained official recognition with the reauthorization of the Higher Education Act (HEA) in 1992, following a bill introduced by Senator Claiborne Pell. This law established a grant program to help HSIs expand their capacity. Amendments in 1998, led by Congressman Rubén Hinojosa, further simplified the definition of HSIs and boosted federal funding. Since 1995, substantial financial support has been directed to HSIs, helping them grow and better serve Latino students.

HSIs form a unique subset of U.S. higher education, representing just 6% of all institutions but enrolling almost half of Latino undergraduates. They are mostly located in areas with high Latino populations, particularly in the Southwest and Puerto Rico. For instance, California, Texas, New Mexico, and Puerto Rico host most HSIs, which tend to be in urban regions, enhancing access for Latino students.

A key feature of HSIs is their affordability and accessibility, providing more opportunities for higher education, especially for low-income and first-generation Latino students. Many HSIs are community colleges and part of public higher education systems, which helps keep costs low. Despite being a small percentage of U.S. colleges, HSIs play a crucial role in the graduation of Latino students.

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## CHAPTER OVERVIEW

### 2: Part II. Finance

2.1: The Evolution of Financial Market Theories- Are Markets Efficient or Not?

2.2: The Heterodox Challenges of Behavioral Finance

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## 2.1: The Evolution of Financial Market Theories- Are Markets Efficient or Not?

**Title:** The Evolution of Financial Market Theories: Are Markets Efficient or Not?

**Author:** Gavin Datta

**Abstract:**

A singular hypothesis to explain the financial markets has been at the forefront of market research for over 60 years. Contrasting opinions regarding the efficiency of financial markets have led to an unsolved debate surrounding human behavior and its potential impacts on the efficiency of markets. This paper reviews the existing literature on both Efficient Markets and Behavioral Finance, concluding with more contemporary literature that attempts to combine the two theories. It summarizes both the theories themselves and the evidence in support of both. By recognizing the effort to reconcile these two contrasting hypotheses with a hybrid market hypothesis, one can conclude that the questions surrounding financial markets remain unanswered. However, the evidence supporting both hypotheses suggests the necessity of a compromise to rationalize these results. Researchers must account for human misbehavior and rational expectations on a spectrum when formulating new market hypotheses, remaining open to changes in market theory as new variables continue to be discovered.

**Keywords:** Efficient Markets, Behavioral Finance, Adaptive Market Hypothesis

### Background

Financial markets come with a long history of attempts at explaining the price movements of assets. The most influential of these theories is the Efficient Market Hypothesis. If one assumes that all market participants act rationally based on the available information, it would be easy to explain all price actions. However, the recurrence of market bubbles, for example, appears to be contrary to the EMH and bears explanation. Behavioral finance directly and increasingly challenges the essential tenets of the EMH by claiming that humans often act non-rationally when making economic and financial decisions. It will remain to be seen whether markets are more or less efficient than non-rational.

### Literature Review

The introduction of the EMH into the academic limelight by Fama (1970) further supports its status as the dominant market hypothesis during his time. His criteria of semi-strong efficiency stated that the price established by the market represented all public information, whereas weak efficiency only considered past price history. This is in contrast to the strong interpretation of the EMH, where all public and private information is included in an asset's price. Through data-driven support of the weak, semi-strong, and strong efficiency interpretations, the EMH developed greater flexibility to adapt to any "anomalies" discovered within the market. The approach by Fama to explain the EMH relied heavily on mathematical analysis.

However, the mathematical approach for proof of the EMH fails to support a fundamental tenet, the Expected Utility Theory. The expected utility theory postulates that rational market actors will always maximize utility. Kahneman and Tversky (1979) explored the idea of human thought processes contrary to the belief of the expected utility theory. Through a survey-style study, they discovered that humans tend to value losses more than they value gains, a behavior named Prospect Theory. Thus, people tend to be more risk-seeking when faced with greater potential losses and more risk-averse when faced with greater potential gains. This research underlined a significant issue with the EMH, as one of its core tenets seemed incorrect. Humans did seem to have consistent irrational biases.

These irrational behaviors were also not limited to just Prospect Theory, as the entire idea of rational behavior by market actors came under question by behavioral scientists. Shiller (1999) explored other recently discovered behavioral tendencies and how they could impact the markets. For example, the Regret Theory states that some market actors don't want to sell stocks that have lost value to finalize their losses but will sell stocks that are higher in price so as not to regret not realizing the gains. Finally, the most important and directly contradictory theory to RET is overconfidence and over/underreaction to information. The theory states that people tend to be overconfident in their interpretation of information, thus over or underreact to information presented in the market. This contradicts the RET, as market participants are expected to act in the best possible manner and react to the best information. So, to claim that market participants could under or overreact would directly contradict the EMH. Nevertheless, Shiller remained somewhat skeptical of the impacts of behavioral finance as a leading theory for market function, as he stresses the importance of proving the widespread impact of these behavioral phenomena.

The main question proposed by Shleifer (2000) leads the discussion surrounding the importance of behavioral psychology in understanding markets. The challenge of applying behavioral psychology to financial markets is that individuals express many

different behaviors. Furthermore, it was always assumed that people who made irrational decisions could be defined as “noise traders,” and it was thought that noise traders were arbitrated away. Understanding whether or not the misbehaviors were correlated is essential in discovering if there was potential for market-wide mispricing.

Fama (1998) still finds fault with the methodology of studies that try to attribute pricing anomalies to widespread misbehavior. Instead, Fama found that markets tended to revert after a period of three to five years. Thus, the most anomalous pricing was due to chance instead of market-wide investor irrationality. Through his statistical analysis of markets, Fama concluded that the methodology used by those attempting to prove irrationality in markets needed to be corrected. Therefore, the notion of market efficiency is still valid.

Nevertheless, with increasing pricing anomalies being observed, researchers were required to figure out the cause. Shefrin (2001) found that investors favored companies with lower book-to-market equity ratios than higher book-to-market equity ratios, even though higher book-to-market equity ratios had higher historical returns. This showed non-rational behavior committed by large portions of investors. However, those supporting the EMH, such as Fama (1993), explained that book-to-market equity was another risk factor to be added to CAPM, explaining the excess returns gained by companies with high book-to-market equity. La Porta (1997) explains that the explanation of the EMH, claiming book-to-market equity is another risk factor, and the explanation of inefficient market advocates may be correct. There is likely some level of preference surrounding risk aversion or preference responsible for the existence of value and growth stocks. It seems that preference does impact the price of stocks, but whether the preference is efficient remains unsolved.

With a more modern acceptance of behavioral finance, Thaler (2005) writes about the importance of understanding how non-rational human behavior impacts the market. He acknowledges the potential of arbitrageurs to correct for mispriced assets but still believes that there could be a widespread error due to common behaviors by market participants. More modern researchers concur with this analysis but interpret the differences between an efficient and inefficient market more flexibly.

Farboodi (2021) found that assets have become more informative as they become larger, whereas smaller cap stocks tend to be less informative. The extent to which assets are informative can indicate efficiency and potentially show that there are degrees of efficiency in the market, even outside Fama's still strict weak, semi-strong, and strong efficiency categories. Cajueiro (2004) also found that markets become more efficient as they develop. It is claimed that deviations from efficiency are due to underdeveloped market status. With the relationship between informativeness and efficiency, developed markets likely see greater efficiency due to higher information transmission and dispersion rates.

Another angle taken to approach the hybrid model of efficiency and inefficiency is in Lo's (2005) introduction of the Adaptive Market Hypothesis. It gives way to both behavioral finance and the EMH, as it marries the two ideas as not mutually exclusive. The theory underlies that people tend to act rationally but can become irrational due to significant volatility in the market. He believes that overconfidence and overreaction are due to factors such as natural selection. Thus, these factors would naturally occur during times of volatility in response to seemingly unexplained price movement. He also adds that people learn from their mistakes, meaning if they try something and it works, they will likely do it again. The AMH argues that investors tend to be rational most of the time but naturally make mistakes, which influences future investment decisions depending on the result of that error. If a decision leads to good results, even if it is irrational and considered a mistake, more of the market would mimic the decision. Which, in turn, could lead to mass market misbehavior. However, if the non-rational decision led to bad investments, the market would move more towards efficiency. Thus, the AMH helps explain why markets can act rationally and create bubbles simultaneously.

Finally, Chu (2019) tested the AMH in high-frequency cryptocurrency markets. Even though there is no established model for testing the AMH, he found periods of efficiency and inefficiency in the market using preexisting methodology to test for such factors. Furthermore, when the market swung from efficiency to inefficiency, it correlated with market volatility, supporting the AMH.

## Analysis

The beginnings of the EMH and further expansion by Fama (1970) didn't initially spell inflexibility by researchers. Evolving from the Random Walk Hypothesis, where it was believed that markets moved randomly and unpredictably, the EMH furthered the ability of investors to understand the root cause of asset prices. More importantly, it gave researchers a practical and straightforward way of projecting market activity and explaining price movements. This, however, shouldn't have been the end of the market hypothesis debates.

When analyzing the behaviors discussed by Kahneman (1979) and Shiller (1999), the evidence is clear that humans have widespread non-rational behavior. Prospect Theory establishes widespread correlated deviance from rational expectations. Furthermore, Regret Theory, Overconfidence, and Over/Underreaction are examples of correlated non-rational behaviors humans express. The studies done in these papers partially answer Shleifer's question about whether humans' non-rational behavior in the market is correlated. Nevertheless, it doesn't fully answer the question about the extent to which non-rational behaviors impact the market.

Even with solid evidence in support of widespread misbehavior, Fama (1998) is mostly successful in defending the EMH by showing how it is still possible that the deviations in the market could be due to chance. This provides a dichotomy of two strongly supported market dynamics hypotheses with no clear solution.

With such a vast disagreement on the nature of markets and strong support for both arguments, the best way to solve the discrepancy between the two theories is to stop viewing them as mutually exclusive. Farboodi (2020) and Cajueiro (2004) both evaluated market efficiency on a spectrum, showing how the market can swing through "levels of efficiency." This idea becomes fully explored through Lo's (2005) AMH. Combining the idea of the EMH on a spectrum with the concepts expressed in behavioral finance creates the perfect compromise between two contrasting ideas.

The evidence supporting both the EMH and behavioral finance concepts shows the necessity for a compromised hypothesis between the two. The existence of a compromise is likely to be the only potential solution, as the existence of pricing anomalies and empirical evidence in support of widespread correlated misbehavior support the concepts of behavioral finance. However, the mathematical support of market efficiency provides strong evidence of efficient markets.

While Lo's AMH might not be the final answer for determining how financial markets work, researchers should be able to somewhat resolve the debate between efficiency and behavioral finance. It isn't a question of either or but should instead be treated as a spectrum.

## Conclusion

Behavioral finance has revolutionized the way we think about the market. While the issue of efficiency versus inefficiency in the market hasn't fully been resolved and likely never will be, accepting that a compromise is the most logical way forward is necessary to advance our understanding of financial markets. Researchers may never discover a perfect hypothesis to explain financial markets, but as new variables explaining market price movements continue to be uncovered, researchers should be wary of treating market hypotheses as orthodoxy.

## Suggestions for Future Research

Defining how a compromise between behavioral finance and the EMH would exist is essential. The AMH provides a potential compromise but lacks a mathematical model to test the hypothesis. Researchers must develop a testable market hypothesis when creating a hybrid hypothesis. Furthermore, exploring the applications of other fields of science to formulate a market hypothesis, as seen through the AMH, could prove valuable in understanding markets. Future research must consider new formulations of behavioral science, efficient markets, and alternative scientific concepts when hypothesizing new market hypotheses.

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## 2.2: The Heterodox Challenges of Behavioral Finance

**Title:** The Heterodox Challenges of Behavioral Finance

**Authors:** Jiaen Peter Guo

**Abstract:** In behavioral finance, it is unclear whether markets are “efficient” or effected by non-rational human behavior. This paper outlines the challenges of behavior to traditional finance and summarizes some of the critical behaviors that individuals manifest.

**Keywords:** Rational Expectations; Efficient Market Hypothesis, Utilitarian Theory, Behavioral Finance, Prospect Theory, Endowment Effect, Bounded rationality

Traditional finance assumes that investors are “rational.” Rationality implies that humans make decisions based on a complete fact set, infinite capacity for mental calculations, and an emotionless mindset. In this paper, the notion of Rational Expectations and its corollary theory, the Efficient Markets Hypothesis, are challenged based on empirical observations of non-rational human behavior.

This paper explores how heterodox behaviors are inconsistent with traditional financial theories and models. In so doing, an unresolved question will be posited having to do with the extent to which these challenges partially or completely negate our understanding of financial behavior and, therefore, valuation models.

### Literature Review

Traditionally, Economics and Finance assumed that market participants at all times make decisions “rationally.” The concepts of “Rational Expectations” and, in general, rationality, assert that individuals look for those opportunities that provide the best results at the lowest possible costs. This concept was first introduced by John F. Muth (Muth, 1961). If we assume that everyone makes decisions rationally, which is also to say that decisions are arrived at in the same manner, then current values for goods and investments would all be the same and outcomes would be virtually identical.

For instance, if people expect the prospective supply of a good to be higher than currently, then people would also expect the price to decrease. Assuming that the individual incorporates rational thinking into his/her decision-making process, he/she may defer his/her purchase to a later time when the supply reaches his/her marketplace and prices have decreased. To make the purchase immediately and not defer his/her purchase until a later date where he/she could potentially obtain a lower price would be considered irrational.

An illustration of the Rational Expectations Theory is depicted in the decision-making process of a high school graduate selecting a college major based on projected earnings in various fields over the forthcoming years. For instance, consider the field of computer science, which was highly lucrative in the early 2000s and continues to be so today. The high demand coupled with a limited supply of coders and software engineers led to exceptionally high salaries, thereby incentivizing more students to declare computer science as their major. Consequently, as the supply of computer science graduates increased and surpassed the demand, salaries began to decline.

In finance, the investor will seek out the highest possible return with the lowest possible risk. Return is profits expressed in percentage terms. More specifically, return is profits divided by the amount of money invested. Risk is a measure of the extent to which the actual return will differ from the expected return. If there is no risk, then actual returns will match the expected returns. (Schoenmaker & Schramade, 2023)

The Efficient Market Hypothesis (EMH) which is based on the work of Eugene Fama (1970), depends on Rational Expectations as applied to financial markets. Once again, as with Rational Expectations, the EMH assumes similar pricing relative to similar investments. Therefore, two securities with similar or identical risk characteristics should be priced similarly. If not, one or the other security would be viewed as being under- or over- priced.

The EMH defines “efficiency” as the accurate pricing of securities based on both return / risk variables, which in turn is based on the information available to investors. It is, after all, information which informs risk versus return points of view. Due to the fact that investors have the same information at the same time and employ the same rational tools to make investment decisions, Fama asserts that it is virtually impossible to consistently earn returns that exceed what would be expected given certain risk parameters. In other words, investors can not consistently “beat the market.”

The efficient market hypothesis (EMH), popularly known as the Random Walk Theory, is the proposition that current stock prices fully reflect available information about the value of the firm, and there is no way to earn excess profits, (more than the market



overall), by using this information (Clarke, 2001). The only way to achieve higher returns would be by taking on higher risk. Higher returns will come only with higher risk.

Utilitarian theory emphasizes maximizing overall benefits derived from rational decision-making. The idea of utilitarian theory, in its original form, considers the balance between pleasure and pain in a purely ethical context. Pleasure was considered a good and pain was considered a bad. (MacAskill, et al., 2023)

In Financial Markets, we may expand on Bentham's notions of good and bad by applying it to risk and return. Return would be considered desirable, a good, while risk would be considered undesirable, a bad. In making any financial decision, investors must consider the balance of return and risk, the good and the bad.

Investors will display different tolerances towards risk. As a result, risk-reward ratios will differ from one investor to another. The differences will determine individual portfolio composition. The theory of Rational Expectations assumes that individuals will act to optimize net personal benefits. This means that investors will choose securities that provide them with the greatest return for any given amount of risk.

Due to the assumption that all investors act rationally, we can now say that aggregate markets reflect the optimization of risk and return. The EMH asserts that aggregate markets acting rationally, consuming information, and using the same tools of analysis will price securities efficiently. "Efficient markets" assumes that risk-return ratios are always optimized. Behavioral finance challenges Rational Expectations Orthodoxy.

It is arguably incorrect to assert that investors, at all times, act rationally in the sense asserted here. Clearly, human beings have emotions, which at times may interfere with rational decision-making processes. Additionally, individuals act without having a complete fact set about matters at hand. Finally, investors are not capable of spontaneously engaging in the highly complex calculations needed to optimize risk-return ratios.

## Behavioral Finance

In this section, we will survey various behaviors that contradict Rational Expectations Theory. Behaviorists, such as Daniel Kahneman and others, have studied the various manners in which people make decisions. Kahneman introduced "Prospect Theory" which asserts that, unlike our understanding of risk in traditional finance, investors value gains and losses unequally.

Prospect theory posits that individuals... give more weight to losses than to comparable gains, and that they are generally risk-averse with respect to gains and risk-acceptant with respect to losses. (Levy, 1992)

In addition to Levy, Kahneman & Tversky made the following comments.

Losses (outcomes below some reference point) are weighted substantially more than gains (outcomes above the reference point) in the evaluation of choice options (Kahneman & Tversky, 1984).

People tend to avoid losses more than to pursue equivalent gains. An investor will fear a prospective loss more than he/she will be encouraged by an equivalent gain.

Another example of irrational behavior is the Endowment Effect. People ascribe more value to things merely because they own them. Ownership does not change the inherent value of the item, but the fear of loss does. An example of this was in 1980, when Dr. Thaler presented half of his Cornell class with coffee mugs, and suggested for them to trade with the latter half of the class, which did not own mugs. Due to the endowment effect, little trading occurred, because owning the mug allowed the students to prescribe the value of the mug unattainable to their counterparts. The half of the class which owned mugs were reluctant to trade their mugs away, even though they would likely have taken little to no interest in owning the mugs, if there was a price listed on them.

Thaler presented half the students in a class with Cornell University coffee mugs and then allowed them to trade with their less fortunate classmates. Surprisingly little trading occurred... Apparently, briefly owning a coffee mug raised its value to the owner sufficiently to price it beyond the reach of most non-owners. (Reb & Connolly, 2023)

The loss in utility associated with giving up a good is greater than the gain in utility associated with getting that good. Or, more simply, losses loom larger than gains.

Though some humans are perceived as "rational", bounded rationality asserts that decision makers intend to be rational, but because of human cognitive capabilities, they sometimes fail on important decisions. Bounded rationality tends to occur when people need to make a decision based on complex factors, people are making decisions based on inaccurate or incomplete

information, people are influenced by emotions that distort judgement, and when people are not able to collect all relevant information due to constraints, such as time or pressure.

The term ‘bounded rationality’ is used to designate rational choice that takes into account the cognitive limitations of the decision-maker — limitations of both knowledge and computational capacity. Bounded rationality ...is deeply concerned with the ways in which the actual decision-making process influences the decisions that are reached. (Eatwell & Newman, 1990)

In addition to Eatwell & Newman, Selten adds the following comment.

Fully rational man is a mythical hero who knows the solutions of all mathematical problems and can immediately perform all computations, regardless of how difficult they are. (Selten, 1999)

Full rationality requires unlimited cognitive capabilities, while humans only have limited cognitive ability. For this reason, the behavior of human beings cannot adhere to the idea of full rationality, and the concept of Rational Expectations.

## Conclusion

This paper has presented just three behaviors that are viewed as non-rational. While there are many more non-rational human foibles than covered in this paper, those that have been presented are sufficient to cast doubt on the validity of our traditional views of finance and financial models.

It is unclear at this time whether markets are completely efficient, only partially, or completely inefficient. It is important to ascertain the extent to which markets reflect efficient prices or not. It is possible that markets are generally, but not always, efficient, as many argue. The answer to this conundrum will profoundly affect the extent to which we accept or reject the traditional valuation models.

## Suggestions for Future Research

Given the numerous challenges posed by behavioral finance, it remains to be seen whether these challenges pose a fatal blow to Rational Expectations and Efficient Market Hypothesis based models. The question regarding the correlation of heterodox behaviors with one another will affect the extent to which security traders will be able, or not be able, to exploit the idiosyncratic “errors” that non-rational players make in the markets arenas. If idiosyncratic behaviors are diverse, then it is conceivable that non-rational actions cancel one another out (Shleifer, 2000). Future researchers may explore the correlation and universality of irrational behaviors as it affects security market valuations.

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## 2.3: The Evolution of Moral Hazard in the United States

**Title:** The Evolution of Moral Hazard in the United States

**Author:** Alice Wang

**Keywords:** Dot.com, Market bubble, Moral Hazard, Long-term Capital Management

### Background

“Moral Hazard” describes excessive spending as a result of being eligible for insurance compensation when losses occur. In this case, the beneficiary is subsidized in the purchase of the goods and therefore continues to consume even after the marginal benefit is lower than the marginal cost” (Marshall 880). Simply put, it is a situation where party A engages in risky behavior knowing that party B will compensate party A for any losses incurred. The concept originated with the advent of private insurance companies about 350 years ago.

Soon after these companies were formed, it was realized that people who purchased insurance were taking risks that they would not have taken without it (Havard).

Nobel Prize winner economist, Paul Krugman, defines moral hazard as:

...a situation in which one person decides how much risk to take, and when things go bad, another person bears the cost (Krugman 63).

Moral hazard has played a notable role in the development of the American economy and its financial systems. This article will explore the development of moral hazard and the future risks it presents based on the three actual cases. It will focus on the failures of the Long Term Capital Management (LTCM) hedge fund in 1998, the Dot.com crisis in 2001, and the banking and mortgage crisis in 2008.

### Analysis

This paper will apply the concept of Moral Hazard chronologically to three salient cases: The near failure of a large hedge fund, and the Dot.com and Mortgage Banking Crises. The reader will note that, in each instance, the notion of Moral Hazard may be reapplied with increasing severity.

### Long-term Capital Management

Hedge funds, such as LTCM, are often referred to as “absolute return funds,” which are expected to generate high returns regardless of the market’s overall direction. These funds are managed by professionals and utilize a variety of investment strategies such as arbitrage, long/short stock trading, futures trading, and high leverage, i.e., borrowed funds (Fleming and Liu).

Hedge funds are usually targeted at high-net-worth individuals and institutional investors because they have a greater tolerance for risk, higher investment capacity, meet qualified investor requirements, are able to cope with liquidity constraints, and have the necessary financial knowledge and experience.

A good example is Long-term Capital Management (LTCM), a hedge fund led by a team of market experts, including Nobel laureates, that was well known for its high-risk, highly leveraged investment strategies. But LTCM faced possible bankruptcy in 1998, and due to its enormous size, it was feared that the hedge fund would severely damage the US’ banking system and the national economy. As a result, the Federal Reserve organized a major economic bailout involving banks and brokerage firms, who prevented LTCM’s assets from being sold off in an already volatile market.

As of the end of 1997, LTCM was holding about thirty dollars in debt for every one dollar of capital. In other words, LTCM was leveraged 30:1.

LTCM returns \$2.7 billion of equity capital to investors (leaving \$4.8 billion of equity capital) while keeping total balance sheet assets near \$125 billion, thus increasing leverage. Debt leverage at LTCM reaches 30 to 1 (Fleming and Liu).

Although high leverage can magnify returns, it also magnifies risk. With very high leverage, very small price fluctuations can lead to huge losses, even to the point of wiping out the entire account.

Secondly, LTCM traded complex derivative instruments, which were high in price volatility and low in liquidity. These factors exacerbated LTCM’s already high leverage-based risk (Fleming and Liu).

In 1997, there was a foreign currency crisis in southeast Asia, starting in Thailand. In 1998, Russia suddenly devalued its currency and defaulted on its debts. These external events led to severe volatility in the global markets, which resulted in significant losses for LTCM's already highly risky portfolio. In August 1998 alone the fund lost 44%. As a result, LTCM faced a possible collapse.

LTCM relied on sophisticated financial models, most notably the "Black-Scholes model." The models, however prestigious, did not incorporate events which were without any historical precedent. In other words, when the 1997-1998 crises occurred the models did not take into account these unpredictable events. The models simply failed, leading to catastrophic losses.

LTCM's financial difficulties underscore the risks associated with high leverage and market volatility. When LTCM's positions faced substantial losses due to severe market fluctuations, the insufficient margin in its accounts failed to meet regulatory requirements. This situation triggered a "margin call," which required LTCM to deposit additional capital that it did not have. Consequently, LTCM's inability to continue operations sparked turmoil in the financial markets (Long-Term Capital Management, 2016).

Furthermore, LTCM's lenders played a crucial role in exacerbating the crisis. The excessive financing provided by the lenders, coupled with their inadequate risk assessment, created a misleading perception of LTCM as a low risk investment. This lack of risk evaluation contributed significantly to the financial instability that followed.

In response to the escalating crisis, LTCM sought intervention from the Federal Reserve. By September 1998, LTCM had reached out to the President of the Federal Reserve Bank of New York to address its financial instability.

The Federal Reserve then orchestrated a rescue plan involving 14 banks, which collectively contributed \$3.625 billion in capital in exchange for a 90% ownership stake in LTCM, leaving LTCM with only a 10% stake and minimizing their financial losses (Fleming and Liu).

This case highlights a significant moral hazard issue inherent in LTCM's approach. The firm's management, composed of Nobel laureates Robert C. Merton and Myron S. Scholes, as well as other renowned financial experts and Wall Street executives, exhibited overconfidence in their complex trading strategies and models. While their theories, such as the Black-Scholes Model, had previously been successful under normal market conditions, they proved inadequate under extreme market conditions.

LTCM's overconfidence led it to assume risks that, in and of themselves, did not presage the possibility of failure. The hedge fund did not imagine the possibility of massive portfolio failure. In its colossal arrogance, the managers held no inkling of what was to come. Thus, there was no ex-ante expectation of a bailout. However, even though there was no direct government involvement aside from moral suasion, market watchers arguably viewed the situation as LTCM's being "too big to allow to fail," and hence opening the door to future moral hazards.

The LTCM case illustrates the broader implications of moral hazard in financial markets. It demonstrates how speculative behavior and overconfidence can lead to severe economic repercussions. To further explore the impact of speculation and its economic consequences, we can examine another significant financial event: the Dot-com crisis.

### The Dot.com Crisis

The "Dot Com Crisis of 2000" was driven by the speculative overvaluation of internet company stocks caused by excessive investor optimism. As the true value of these companies became apparent, a market correction ensued, precipitating a broader market collapse.

A peculiar feature of the "dot-com" boom was that it began in the late 1990s in the most liquid and diversified market in history. Market participants continued to artificially enhance the boom, despite numerous warnings about the inflated rates at which investors bought shares of advertised companies of the so-called "new economy." When investors saw the huge profits obtained by the first buyers of Microsoft and several other major IT-corporations, they believed that any company associated with computers or the Internet will give the same profit. Total boom around information technologies was at a maximum at that time, and therefore the revenues that could be obtained from the use of the Internet in business significantly revalued. (S.Z. Moshenskyi and Сергій Захарович Мошенський).

Thus, an important factor in the formation of the internet crisis was the emergence of a large number of new internet companies. Many of these companies hoped to obtain capital by issuing shares without considering whether selling goods and services over the internet would be profitable. Financial analysts focused on price to sales rather than price to earnings ratios. In 1999, the information technology boom reached its peak.

The boom reached its peak at the end of 1999 i.e. the NASDAQ stock exchange index, where shares of most Internet companies were quoted, nearly doubled during the year, as the Dow Jones index by 20 % (S.Z. Moshenskyi and Сергій Захарович Мошенський).

As a result of this tech bubble crash many companies ceased to exist. The dotcom bubble started collapsing in 1999, and the fall precipitated from March 2000 until 2002. Several tech companies that conducted an IPO during the era declared bankruptcy or were acquired by other companies (CFI Team).

Such a wild stock rally greatly overestimated the actual and potential profits internet companies offered. This bubble-like stock market could not be sustained and stock prices declined rapidly from 2000 onwards.

The NASDAQ index lost half of its value between March and December 2000, while the S&P 500 index lost 14 %.The losses were huge. The impact of the rapid decline in stock prices could have been huge financial losses for investors. The losses were huge (S.Z. Moshenskyi and Сергій Захарович Мошенський).

This is especially true for those who purchased stocks at the peak of the boom. The value of their investments diminished sharply. Many companies went bankrupt.

By the end of 2001, most dot-com stocks like pets.com, Webvan, etoys, and Kozmo went bust, although some companies survived (Terrell).

The fact that so many people overvalued internet company stocks was a remarkable phenomenon. The eagerness to finance the initial public offerings (IPOs) of Internet startups led to an unsustainable surge in stock prices. Venture capitalists' eagerness to fund IPOs of internet startups contributed to the boom. This influx of capital created an overheated investment environment, magnifying the speculative frenzy and inflating the dot-com bubble to unsustainable levels (Duignan).

Lastly, accounting regulatory issues compounded the crisis. Many internet companies during this period issued inaccurate or exaggerated financial statements, obscuring their true financial health and performance.

As the valuations of shares in new and existing dot-com companies continued to rise, many investors became convinced that the U.S. economy had been fundamentally transformed and that several factors that had traditionally figured in the valuation of a company's shares—such as the assets and liabilities on its balance sheet, the revenue and profit on its income statement, as well as market share and cash flow—were not directly relevant to assessing the future performance of dot-com companies, particularly start-ups (Duignan).

This lack of transparency made it challenging for investors to accurately assess risks, heightening investment uncertainty and contributing to market volatility.

In summary, speculation, the irresponsible use of technical analysis, and venture capitalists' excitement about internet start-up IPOs all contributed to the dot-com bubble. The over-injection of capital led to unsustainably high stock prices. Many internet companies also issued inaccurate financial statements, making it hard for investors to assess risks. To avoid similar issues in the future, it is crucial to be vigilant about speculative bubbles and to strictly enforce accounting regulations.

### The Mortgage Banking Crisis

The mortgage banking crisis centered around adjustable-rate mortgages (ARM). A mortgage is a loan provided to finance residential home purchases. An ARM is a type of loan where the interest rate adjusts based on changes in an index.

Typically, this adjustment results in an increase or decrease in the monthly payment amount of the loan (Financial, Consumer, and Protection Bureau 10). ARMs have an initial period after which the interest rate is adjusted on a regular basis. ARMs attracted many homebuyers by offering a low, short-term rate in the early years. However, the risk is that the repayment is uncertain, and if interest rates rise after the initial fixed rate period, the monthly repayment may increase significantly (Barberis).

Additionally, the low interest rate made the mortgage appear affordable to the borrower who was not aware that after a period of usually 5 years the mortgage rate would reset to a fixed and much higher long-term rate that was no longer affordable.

As interest rates on ARMs rose in line with market rates, many borrowers were unable to afford their monthly payments. This led to widespread defaults and foreclosures, creating an oversupply in the housing market that caused residential real estate prices to plummet. The correction in the housing industry triggered an overall economic downturn.

The adjustment in the real estate sector also led to a downturn in consumption in the U.S. economy, which in turn triggered a downturn in the real economy, creating a cycle of economic decline (Li 67).The lending crisis has led to an overall economic

downturn that has severely weakened the global financial system. This is specifically reflected in the fact that \$7.4 trillion in stock market value and \$3.4 trillion in real estate wealth were wiped out. Many companies went bankrupt, and 7.5 million Americans lost their jobs, causing the unemployment rate to rise from 5% to 10% by 2010. Although jobs started to return, many were lower-paying and less secure. Overall, American households saw their net worth fall by \$17 trillion, a 26% drop (Chang).

In the wake of the crisis, there has been a significant slowdown in global economic growth. Business investment declined and consumer spending was curtailed, leading to a slow recovery process. In response to the crisis, many countries have strengthened financial regulation and introduced new policies and regulations to prevent the recurrence of similar events. These reforms include increased scrutiny of financial institutions, greater transparency and improved consumer protection measures.

As a result of the [recession](#), Congress responded by passing multiple laws to help prevent another financial crisis from happening again. They passed the Dodd-Frank legislation, which included the Mortgage Act and the Consumer Financial Protection Act (Chang).

In summary, adjustable-rate mortgages revealed serious economic impacts and risks during the financial crisis. Future economic development needs to focus on continued policy adjustments and market stabilization, along with enhanced regulation and monitoring. The global financial system needs to be continuously improved and promote sustainable economic growth.

### Summary

The examination of moral hazard through the lens of the Long-Term Capital Management (LTCM) failure, the Dot-com Crisis, and the Mortgage Banking Crisis reveals critical insights into the complexities of financial risk in the U.S. economy. Each case illustrates how the interplay of overconfidence, speculative behavior, and inadequate regulatory oversight can precipitate significant economic turmoil.

The LTCM case exemplifies how reliance on complex financial models without considering unforeseen events exemplifies moral hazard. The resulting bailout not only prevented immediate disaster but also raised concerns about the "too big to allow to fail" phenomenon, setting a precedent for future risks.

The Dot-com Crisis illustrated the perils of speculative investments in a rapidly changing market. Misguided optimism led to unsustainable valuations, ultimately causing a market correction that affected numerous firms. This underscores the need for transparency and careful evaluation of investment practices.

The Mortgage Banking Crisis revealed the systemic risks of adjustable-rate mortgages and the lack of borrower understanding regarding interest rate fluctuations. The resulting defaults caused widespread financial disruption, highlighting how misinformed individual decisions can ripple through the economy.

In summary, these three cases elucidate the evolution of moral hazard, underscoring the critical importance of regulation and risk management in financial markets. Future research and policymaking should prioritize the effective identification and mitigation of moral hazard to foster sustainable economic growth and financial stability. Addressing this issue necessitates interdisciplinary collaboration, particularly among the fields of economics, finance, and policymaking, to ensure that we learn from past lessons and prevent the recurrence of similar crises.

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## 2.4: Beyond the Efficient Frontier- Transactional Anomalies and Behavioral Biases Across Markets

**Title:** Beyond the Efficient Frontier: Transactional Anomalies and Behavioral Biases Across Markets

**Author:** Xinao He

**Abstract:** Rational expectations theory, efficient market hypothesis and behavioral economics provide different perspectives for market behavior and prediction. However, the existing research on the influencing factors of stock price changes under different stock trading systems is still very scarce. This paper constructs an economic model that integrates rational expectations theory, efficient market hypothesis and behavioral economics, and uses vector autoregression (VAR) and error correction model (ECM) to empirically analyze the trading data of Chinese and American stock markets. The results show that : (1) The introduction of behavioral economics has unique advantages in explaining market overreaction and adjustment lag, which provides a powerful supplement to rational expectations theory and efficient market hypothesis. (2) The degree of information disclosure in the stock market and the deviation of investors' behavior have a significant impact on stock price volatility and market adjustment efficiency. (3) Compared with the US stock market, the Chinese stock market is more susceptible to investor behavior bias. The study also highlights the importance of combining rational and irrational factors in formulating economic policies, and puts forward the corresponding policy suggestions.

**Key words:** Rational expectations theory; Efficient market hypothesis; Behavioral economics; Comparative analysis; Behavior of the market

With the development of market economy and the complexity of financial market, it has been difficult for traditional economic models to explain all market phenomena alone. As a result, new models and approaches that integrate these existing theories are becoming the focus of attention in both academia and industry. While rational expectations theory emphasizes the rationality of individual expected decisions under complete information, the efficient market hypothesis argues that financial markets can quickly and efficiently reflect all relevant information. However, behavioral economics reveals systematic irrational behavior in the human decision-making process, and the combination of these theories provides a more comprehensive perspective for analyzing and predicting market behavior. An empirical study will be conducted to explore the practical applicability of these theories in the current economic environment and their potential value in forecasting and strategic development. In addition, examining how the combination of these theories and their interactions affect specific economic strategies and decisions, especially their application in financial markets, will provide strong theoretical support and practical guidance for investors and market institutions.

The United States and China are not only the two largest economies in the world, but also the largest developed countries and the largest developing countries in the world, and their stock markets are naturally representative. Since market prices are adjusted by behavioral biases on the basis of rational expected prices, we need to understand more clearly the differences in the effects of behavioral biases in different market mechanisms. The purpose of this study is not only theoretical discussion, but more importantly, to verify the practical utility of the theory through specific economic models and econometric analysis, so as to provide references for scholars to study the characteristics of the stock market at different levels of development and policy makers to improve policy content.

### 2.1 Rational Expectations Theory

Rational Expectations Theory (RET) holds that individuals are rational and use all available information to make fair, informed predictions about the future. It pointed out that individuals do not make systematic errors in their predictions and their predictions are not affected by past errors (Muth, 1961). Government policies (such as changes in monetary or fiscal policy) may not be as effective if individual expectations are not taken into account. It also clarified the relationship between rational expectations and monetary policy (Lucas, 1972). That is, when individuals hold rational expectations, changes in money supply will not have an actual impact on the economy, and money neutrality remains unchanged. And by assuming that individuals are forward-looking and rational, rational expectations can help explain fluctuations in key economic variables such as real interest rates and natural employment rates.

### 2.2 Efficient Market Hypothesis

The Efficient Market Hypothesis (EMH) asserts that asset prices in a market already fully reflect all relevant information. According to the research, different forms of this hypothesis (weak, semi-strong and strong) have a fundamental impact on the formulation of asset pricing models and investment strategies (Suciu, 2015). However, behavioral finance theory has questioned

this, pointing out that the decision-making process of market participants is not completely rational, and the market arbitrage behavior is limited.

### 2.3 Behavioral Economics in Financial Transactions

Behavioral economics explains the irrational behavior observed in markets by introducing psychological principles. Gurgenzidze (2024) pointed out that although the traditional economic model assumes that economic actors are rational, the behavior in the actual market often deviates from this preset, showing obvious psychological and emotional effects. In addition, the study by Xie (2023) and Liu (2023) further explored the application of behavioral finance in quantitative investment models, showing that incorporating psychological and behavioral factors into economic models can improve the explanatory power and the forecasting accuracy of the models.

Based on the comprehensive analysis of the existing literature, it can be seen that while rational expectations theory and efficient market hypothesis provide a strong theoretical foundation for economic models, the introduction of behavioral economics adds the necessary complexity and realism to these models. The research of Li (2022) revealed the limitations of the hypothesis of economic man in the real economic environment, especially the application of the hypothesis in China's stock market and emphasized the impact of irrational behavior on market efficiency. Through a comprehensive review of this literature, this chapter not only clarifies the theoretical and practical implications of each theory, but also points out the shortcomings of existing research and possible directions for future research, especially in how to apply these theoretical fusions to a wider and complex market environment.

In exploring how Rational Expectations Theory, Efficient Market Hypothesis, and Behavioral Economics interact and affect market behavior, it is crucial to develop a reasonable theoretical model. The purpose of this chapter is to construct a comprehensive model that integrates the key elements of the three theoretical frameworks described above and proposes a series of hypotheses that can be empirically tested.

#### 3.1 Construction of Theoretical Model

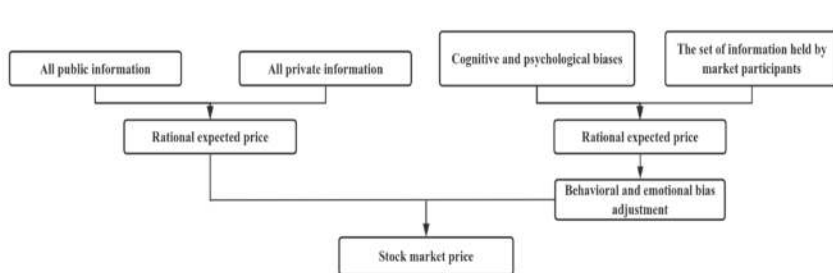
Considering the complexity of the real economic environment, traditional economic models often cannot explain all market phenomena by themselves. Therefore, this study proposes a fusion model, which aims to provide a more comprehensive market analysis framework by integrating the decision efficiency of rational expectations theory, the information processing power of efficient market hypothesis, and the irrational behavior factors in behavioral economics. The model assumes that market participants have different sets of information, including public and private information. Public information is widely shared and easily accessible, while private information varies from individual to individual.

According to the rational expectations theory, if investors can obtain all the market information, then the expected price is equal to the rational expected price, but in fact the information obtained by market participants is not complete. Market participants will reasonably formulate expectations about future market conditions based on the information set they hold.

At this time, the expected price of market participants deviates from the rational expected price, which will lead to the adjustment of the actual market price and the formation of the final market price. In addition, the concepts of behavioral economics are introduced to account for systemic biases in decision-making, such as overconfidence, anchoring effects, and herd behavior. The market's price adjustment mechanism is based on the efficient market hypothesis, which states that prices quickly and accurately reflect changes in all relevant information. However, due to the existence of behavioral bias, the speed and accuracy of market response can be affected.

By comprehensively considering the interaction between rational expectation prices and behavioral deviations that trigger price adjustment, we obtain an analytical framework for the stock market price formation mechanism, as shown in Figure 1:

**Fig. 1** Stock market price analysis framework



Similarly, if we set  $P_t$  to be the market price at time  $t$ ,  $E_t$  to be the rational expected price based on the information set  $I_t$ , and  $B_t$  to represent the price adjustment due to behavioral deviations, then the formula for calculating the market price can be expressed as follows:

$$P_t = E_t(I_t) + B_t(\delta)$$

### 3.2 Hypothesis Establishment

Based on the above model, this study establishes the following hypotheses for verification through subsequent empirical analysis:

**Hypothesis 1:** The effect of rational expectations is more significant in market environments where information transparency is high and widely distributed.

**Hypothesis 2:** Especially in an environment of high market pressure and asymmetric information, behavioral biases will significantly affect price formation in the short run.

**Hypothesis 3:** In an ideal efficient market, price adjustments should reflect all information quickly and without bias, but behavioral biases and imperfect information can lead to inadequate market adjustment or overreaction.

### 3.3 Selection and Application of Econometric Methods

In order to test the above hypotheses, this study will adopt a variety of econometric methods, including regression analysis, panel data analysis and time series analysis. This study pays special attention to the use of vector autoregressive (VAR) models and error correction models (ECM), which can effectively analyze the short and long run effects of information changes, expectation adjustments and behavioral deviations on market prices.

The vector autoregressive model (VAR model) is a commonly used econometric model, proposed by Christopher Sims in 1980. The VAR model uses all the current variables in the model to perform regression on several lagging variables of all the variables. VAR model is used to estimate the dynamic relationship of the joint endogenous variables without any prior constraints. It is a generalization of the AR model, which has been widely used. It was also used to investigate the dynamic relationship between different time series variables and analyze how market information and behavioral deviations affect each other over time.

Davidson et al. (1978) proposed an error correction model. First of all, the cointegration analysis of variables is carried out to find the cointegration relationship between variables, that is, the long-term equilibrium relationship, and this relationship is used to construct the error correction term. Then a short-term model is established, and the error correction term is regarded as an explanatory variable. Together with other explanatory variables reflecting short-term fluctuations, a short-term model is established, that is, an error correction model. The error correction model is used to capture the adjustment process after the short-term deviation from the long-run equilibrium and evaluate how the market responds to the conflict between rational expectations and irrational behavior.

## 4. Empirical Analysis

This chapter aims to verify the theoretical models and hypotheses established in Chapter 3 through specific empirical analysis. A variety of econometric models have been employed to explore how rational expectations theory, the efficient market hypothesis, and behavioral economics together influence market behavior and price formation.

### 4.1 Data Sources and Variable Description

The first step in the empirical analysis is to collect and process the data. In the first part, this study chooses the daily trading data of the US stock market as the object of analysis, and the time range is from 2010 to 2020. The data include stock prices, trading volume, market news reports, macroeconomic indicators and other information, which mainly come from Bloomberg, Yahoo Finance and BW index. To ensure the accuracy and consistency of the data, the collected data were carefully pre-processed, including data cleaning (removal of missing and outliers), variable transformation (such as log transformation), and normalization. In addition, a sliding window approach is used to calculate various statistical descriptions, such as mean, standard deviation, etc. to facilitate the subsequent dynamic analysis. Table 1 indicates the definitions, symbols, measurement units, and sources of model variables.

**Table 1.** Variable Description

| Variables | Acronyms | Status | Measurement Unit | Source |
|-----------|----------|--------|------------------|--------|
|           |          |        |                  |        |

|                                  |     |             |                         |               |
|----------------------------------|-----|-------------|-------------------------|---------------|
| Adjusted closing price of shares | APS | Dependent   | Millions                | Yahoo Finance |
| Trading volume                   | TV  | Independent | Millions                | Yahoo Finance |
| News Impact Index                | NII | Independent | Cumulative Index (-1,1) | BW index      |
| Behavioral Bias Index            | BBI | Dependent   | Cumulative Index (-1,1) | BW index      |
| Standardized stock price         | SSP | Moderating  | Standardized treatment  | Bloomberg     |
| Standardized trading volume      | STV | Control     | Standardized treatment  | Bloomberg     |

Note: BW index: It is a composite index of investor sentiment proposed by Baker and Wurgler (2006)

In the second section, this study mainly discusses the Chinese A-share market. We developed a PCA based composite behavioral deviation index for investors, and developed a composite index to measure the behavioral deviation of Chinese investors by taking the Shanghai and Shenzhen A-share markets as the research object. Five indicators, namely consumer information index, consumer price index, the difference between M1 and M2, stock market turnover rate and price-earnings ratio, are selected as proxy variables of investor behavior, which can better reflect the behavioral deviation of investors in the market. We further refine this research by updating the trading data to be consistent with the time horizon of the trading data in the first part of the study and dealing with outliers in it. The main data sources include the National Bureau of Statistics of China, the People's Bank of China, and the RESSET database.

## 4.2 Model Estimation and Result Analysis

### 4.2.1 The United States Stock Market

#### 4.2.1.1 Use Vector Autoregressive (VAR) Model

First, the vector autoregressive (VAR) model is used to analyze how information sets, market expectations and behavioral biases affect each other in time series. VAR model includes market information indicators (such as market news impact index), market expectation (estimated by historical price trend) and behavioral deviation indicators (such as irrational behavior index obtained by investor sentiment analysis). The estimation results of the model show that market information has a significant immediate impact on price formation, while the impact of behavioral bias is more lagged and variable. Specifically, when market information increases, prices are expected to adjust quickly, but price fluctuations due to behavioral biases become apparent only several trading days later.

**Table 2.** Summary of vector autoregressive (VAR) model results

| Project                                   | Description |
|-------------------------------------------|-------------|
| BIC (Bavesian information criterion)      | 0.0640      |
| HOIC (Hannan-Quinn information criterion) | 0.0374      |
| AIC (Akaike information criterion)        | 0.0224      |
| Likelihood function                       | -16295.9    |
| FPE (The variance of the forecast error)  | 1.0227      |

The immediate impact coefficient of news impact index on stock price is  $-0.0249$  (t-statistic is  $-1.389$ , p value is  $0.017$ ), which means that the short-term impact of news on stock price is significant. And the effect of the behavioral deviation index is not significant (coefficient is  $-0.0048$ , p-value is  $0.806$ ). The influence coefficient of behavioral deviation index on trading volume is  $-0.0439$  (t-statistic is  $-2.181$ , p value is  $0.029$ ), indicating that behavioral deviation has a significant impact on trading volume. The lagged response of stock price to news impact is not significant (coefficient is  $-0.0033$ , p value is  $0.865$ ). Neither stock price nor trading volume has a significant effect on the behavioral bias index. The low residual correlation among the variables indicates that the model is relatively independent and may have a more complex dynamic relationship.

**Table 3.** Coefficient results of each equation

| Variable                                                  | Coefficient | t-Statistics | Prob. | Description                                                   |
|-----------------------------------------------------------|-------------|--------------|-------|---------------------------------------------------------------|
| Standardized stock price<br>(News impact index)           | -0.0249     | -1.389       | 0.017 | The short run effect of news on stock prices is significant   |
| Standardized stock price<br>(Behavioral deviation index)  | -0.0048     | -            | 0.806 | The effect of behavioral bias is not significant              |
| Standardized trading volume (Behavioral bias index)       | -0.0439     | -2.181       | 0.029 | Behavioral bias has a significant impact on trading volume    |
| News impact index<br>(Stock price)                        | -0.0033     | -            | 0.865 | The response of lags to the impact of news is not significant |
| Behavioral Bias Index<br>(Stock price and trading volume) | -           | -            | -     | Non-significant                                               |

#### 4.2.1.2 Error Correction Model (ECM)

In order to further explore the adjustment process of the market after the deviation from the long-run equilibrium, the error correction model (ECM) is used. The model helps analyze how short-term market movements adjust to restore equilibrium in the long run. The results show that behavioral bias plays an important role in the adjustment process after market overreaction. Specifically, when the market deviates from the equilibrium state due to irrational behavior, the speed of adjustment back to the equilibrium state is slow, which indicates that the efficiency of the market mechanism is affected by the behavioral deviation.

Error correction model (ECM) results show that a long-run equilibrium relationship is assumed to exist. The lag effect of standardized stock prices on themselves is significantly negatively correlated with a coefficient of -0.1869, indicating that past prices have a negative impact on future price changes. The positive correlation coefficient of trading volume is 0.2577, indicating that the increase of trading volume is positively correlated with stock price.

**Table 4.** Overview of error correction model (ECM) results

| Project                 | Description                                  |
|-------------------------|----------------------------------------------|
| Lag order               | 1                                            |
| Level of co-integration | There is a long-run equilibrium relationship |

**Table 5.** Coefficient results and error correction terms for each equation

| Variable                                              | Coefficient | Description                                                  |
|-------------------------------------------------------|-------------|--------------------------------------------------------------|
| Standardized stock price (Lagged effect on itself)    | -0.1869     | Past prices have a negative impact on future price movements |
| Standardized trading volume (Lagged effect on itself) | -0.2996     | Negative correlation is significant                          |
| News Impact Index (Stock price)                       | 0.0882      | Positive correlation is significant                          |
| Behavioral deviation index (Lagged effect on itself)  | -           | Non-significant                                              |

**Table 6.** Error correction rate (Alpha)

| Variable                    | Coefficient | Description                                                                          |
|-----------------------------|-------------|--------------------------------------------------------------------------------------|
| Standardized stock price    | -0.6093     | The market adjusts back to equilibrium more quickly after deviating from equilibrium |
| Standardized trading volume | -0.5079     | The adjustment is faster                                                             |
| News impact index           | -0.1713     | The adjustment is slower                                                             |
| Behavioral bias index       | -0.0125     | Non-significant                                                                      |

#### Note

Cointegration relationship: The cointegration vector shows a long-run equilibrium relationship among stock price, trading volume, news impact index, and behavioral deviation index, with stock price and trading volume being the most closely related

### 4.2.2 China Stock Market

In this section, the VAR model will be used to describe the relationship between investors' behavioral deviations and the time series of the stock market index. We establish a binary VAR model, which takes the SSE Composite index and investor sentiment index as dependent variables to establish their own autoregressive linear functions, and the explanatory variables are composed of lagged terms of dependent variables and lagged terms of independent variables. In order to test the impact of behavioral bias on the Shanghai stock market, this paper considers the following regression equation:

$$SI_t = a + \beta_1 SI_{t-1} + \beta_2 SI_{t-2} + \theta_1 BDI_{t-1} + \theta_2 BDI_{t-2} + \varepsilon_t$$

Where SI represents Shanghai Composite Index; BDI stands for investor behavioral Deviation index.  $a$ 、 $\beta_1$ 、 $\beta_2$ 、 $\theta_1$ 、 $\theta_2$  are the band estimated parameters. In the above equations, due to the strong autocorrelation between the deviation of investor behavior and the SSE index, the first-order and second-order lagged variables of investor sentiment are introduced into the regression equation. We use the AIC information criterion and the LR test to select the number of subsequent periods of the VAR model.

**Table 7.** VAR model lag period selection

| Lag | LogL      | LR        | FPE       | AIC       | SC        | HQ        |
|-----|-----------|-----------|-----------|-----------|-----------|-----------|
| 0   | -82.76438 | NA        | 0.347858  | 4.537834  | 4.609832  | 4.556325  |
| 1   | -18.87346 | 118.8574  | 0.012343  | 1.293496  | 1.564312  | 1.383463  |
| 2   | -10.23489 | 14.74832* | 0.103281* | 1.093284* | 1.683829* | 1.245480* |
| 3   | -6.385762 | 6.092384  | 1.108766  | 1.107653  | 1.797273  | 1.324556  |

#### Note

\* indicates the optimal lag order of the model that should be chosen under the corresponding criteria

The results shown in Table 7 show that the VAR model in this section should be chosen to be lagged by 2 periods. Based on the above analysis, a binary VAR model of investors' behavioral deviation index BDI and Shanghai Stock Index SI should be established. The results of the model are shown in Table 8.

**Table 8.** Estimation results of VAR

|          | BDI                   | SI                      |
|----------|-----------------------|-------------------------|
| BDI (-1) | 0.643525<br>[3.87659] | 0.170452<br>[1.04673]   |
| BDI (-2) | 0.312098<br>[1.72534] | -0.002356<br>[-0.01557] |



|                |                         |                         |
|----------------|-------------------------|-------------------------|
| SI (-1)        | 0.427382<br>[1.99034]   | 1.323468<br>[7.23421]   |
| SI (-2)        | -0.456732<br>[-2.50132] | -0.533654<br>[-3.16054] |
| C              | 0.059031<br>[1.16289]   | 0.002531<br>[0.04563]   |
| Adj. R-squared | 0.916745                | 0.9226593               |

In view of the fact that the influence of investors' behavioral deviation on the SSE index is greater than the influence of the SSE index on investor sentiment, it is reasonable to use the lagged values of investor sentiment and the SSE index to explain the SSE index, and the following results are obtained:

$$SI = 1.3235SI(-1) - 0.5337SI(-2) - 0.1705BDI(-1) + 0.0024BDI(-2) + 0.0025$$

$$t = (7.23421, -3.16054, 1.04673, -0.1557, 0.04563)$$

Since the values of BDI and  $\varepsilon_t$  are very small, and the value of t test is not significant, in order to express the relationship between the two more obviously, we decided to remove these two variables, and finally obtained the relationship between investors' behavioral deviation and Shanghai Composite Index as follows:

$$SI = 1.3235SI(-1) - 0.5337SI(-2) - 0.1705BDI(-1)$$

The regression results show that investors' behavioral deviation has a significant impact on the Shanghai Composite Index, but its impact has a lagged time difference, and its lagged order value cannot make investors' sentiment bring a significant positive effect on the Shanghai Composite index. Although the stock price is volatile, the impact of investor behavior deviation on the stock price index is obvious. Compared with the content of Section 1, it is easy to find that compared with the US stock market, investors' behavioral deviations and emotional changes in China's A-share market have a more significant impact on the stock price index.

## 5. Discussion and Policy Implications

This study builds and verifies a comprehensive economic model to explain market behavior by integrating rational expectations theory, efficient market hypothesis and behavioral economics. The empirical analysis shows that although rational expectations and efficient market hypothesis provide the basic explanation for market behavior, the introduction of behavioral economics can explain the irrational phenomena in the market more comprehensively, such as price overreaction and adjustment delay.

### 5.1 Discussion

Through the analysis of VAR model and ECM, it can be observed that rational expectations and efficient market hypothesis have strong support in the data, but behavioral economic theory provides additional explanatory power, especially in explaining the market's overreaction and bias adjustment process. These findings emphasize that in modern financial markets, a single theoretical framework is often insufficient to explain the complexity of market behavior, while a comprehensive theoretical model, taking into account the joint role of rational and irrational factors, can reveal the market dynamics more comprehensively. The empirical results also point to possible directions for future research, including how to quantify behavioral biases more precisely, how to incorporate more types of information in the model, and how to cope with the effects of information updates and changing market conditions.

The results highlight the importance of considering the irrationality of market behavior in economic policy making. Irrational behaviors in the market, especially deviations in the process of information processing and expectation formation, have a profound impact on market efficiency and policy implementation effects. For example, overreaction of the market in the face of major economic information or policy changes may lead to inefficient resource allocation, and policy makers should take more account of these irrational factors when designing interventions.

At the same time, our empirical research finds that compared with the US stock market, China's A-share market is more vulnerable to the influence of investors' behavioral deviations, especially mood changes, which may be related to the lack of investor education, the immaturity of investors' decision-making mechanism, and the crisis of confidence in the stock market and information disclosure mechanism. The occurrence of behavioral deviation will significantly reduce the efficiency of the market mechanism, thus affecting the investment return of small and medium investors in the long run, and starting a new round of vicious

cycle. In China's A-share market, the influence of the investor behavior deviation index on the Shanghai Composite Index is significantly greater than that of the Shanghai Composite Index on the investor behavior deviation index, which also implies that the investor behavior may be more affected by the impact of external information and subjective decision-making.

Our conclusion shows that China's stock market is more susceptible to the impact of investor behavior bias, which means that China's stock market is more susceptible to changes in market information, changes in fiscal and monetary policies, and the trend of global stock markets when there are no significant changes in the operating conditions and valuation of listed companies. When the overall operating conditions of listed companies are relatively deteriorating, investor confidence has become a key factor driving up the stock market. This can cause wild swings in stock market indices in the short term.

## 5.2 Policy Implications

Improving the transparency and accessibility of market information can reduce information asymmetry and market overreaction. Policy makers should strengthen the information disclosure system to ensure that all market participants have timely access to important information. In addition, enhancing investors' financial education and understanding of behavioral deviations can improve the rational behavior of market participants, thus improving the overall efficiency of the market. Regulators should develop more detailed market monitoring mechanisms to adjust market rules and prevent potential market abuses. When formulating economic policies, it is crucial to account for behavioral factors in the policy transmission process to ensure that policy effects are not undermined by irrational market reactions. Especially in the immature stock market of developing countries, it is easy to be affected by the deviation of investors' behavior.

## 6. Conclusion

This paper constructs a model that integrates rational expectations theory, efficient market hypothesis and behavioral economics, and analyzes the interactions and limitations of these theories in explaining market behavior. The empirical results not only support the basic ideas of rational expectations theory and efficient market hypothesis but also reveal the importance of behavioral economics in explaining market anomalies. The comprehensiveness of the model enables it to predict market behavior more accurately and provides richer theoretical support for policy making. This study provides important implications for the improvement of financial market regulatory policies, investor education and information disclosure system. The study highlights the need to take rational and irrational factors into account in economic models and policy making. Future research could further explore the applicability of these theories in different markets and regions, and how market structure affects the predictive accuracy of theories.

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